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Where will growth come from?

Reallocation, the missing factor in Hungary

Abstract

Hungary needs growth that does not burden the public budget. We ask whether part of it can come from reallocation: workers and capital moving from less to more productive firms, including through the entry and exit of firms. Using balance sheets of all Hungarian double-entry firms for 2004–2023 and CompNet data for Poland and other Central and Eastern European countries, we run three exercises. First, we compare productivity gaps between firms. By the late 2010s these gaps, within the same industry, were wider in Hungary than in Poland and the region on every measure we have, and in Hungary they did not narrow after 2019. Second, we decompose productivity growth. The contribution of reallocation fell by almost half after 2013, from 1.6 to 0.8 percentage points a year. About two-thirds of this fall comes from foreign-owned firms: they stopped gaining employment share from less productive domestic firms, and reallocation among them weakened. Among domestic firms reallocation slowed by about a quarter. Third, we use a standard misallocation model. Its efficiency gap, the output that better allocation could add, widened among domestic firms at least as much as overall, most in construction and utilities. Returning to the allocation of 2004–2010 would raise the level of output by about 1–2% of GDP: a one-off but lasting gain, or 0.2–0.4 percentage points of extra growth a year over five years. Rules that apply equally to all firms, time-limited sector measures, open entry and exit, and transparent support could mobilise this reserve; the state choosing which firms should grow would not.

Keywords: productivity, reallocation, misallocation, firm dynamics, Hungary.
JEL: D24, L25, O47, O52.

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Executive summary

The message. Hungary needs to reduce a large budget deficit, expected at 7.5% of GDP in 2026, and is looking for growth that does not cost the budget. One such source is already in the economy. Aggregate productivity grows when firms improve, and also when workers and capital move from less productive to more productive firms, including through entry and exit. In Hungary this second channel, reallocation, has weakened, and productivity gaps between firms did not narrow as they did in Poland and the region. Economic policy that gives firms more room to grow, shrink, enter and exit could mobilise this reserve.

Growth that does not cost the budget: a reserve in how resources are allocated

What we found (balance sheets of all Hungarian double-entry firms, 2004–2023; CompNet data for Poland and the CEE region):

- **Gaps between firms stayed wide.** By the late 2010s, productivity gaps between firms in the same industry were wider in Hungary than in Poland and the CEE region on every measure we have, by about 0.1 on a spread of around 0.7. On our main measure, Poland's gaps were close to those of the old EU member states. In Hungary they did not narrow after 2019.

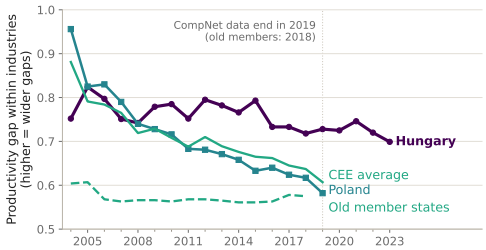
Exercise 1: Hungary's gaps about 0.1 wider than Poland's, on every measure

- **Resources moved less.** Workers moving to more productive firms, and weak firms exiting, added 1.6 percentage points a year to productivity growth in 2004–2013 and 0.8 in 2013–2022, a fall of almost half. About two-thirds of the fall comes from foreign-owned firms, which stopped expanding at the expense of less productive domestic firms. Since 2013, productivity growth has come mostly from firms improving themselves.

Exercise 2: from 1.6 to 0.8 points of growth a year

- **The reserve is sizeable.** If allocation only returned to its 2004–2010 state, a standard model suggests that the same workers and capital would produce about 1–2% of GDP more: a one-off rise in the level of output that then persists. Reached over five years, this is 0.2–0.4 percentage points of extra growth a year.

Exercise 3: 1–2% of GDP, a one-off rise in output



Spread of firm TFP (standard deviation over mean) within two-digit industries, firms with 20+ employees; higher = wider gaps. Hungary: own data. Poland, CEE average (2004 joiners, incl. Hungary) and old member states (nine EU-15 countries): CompNet, to 2019.

What this suggests. Reallocation works when markets allow productive firms to grow and weak ones to shrink or exit. The research on misallocation points to rules that apply equally to all firms, open entry and competition, sector-specific measures that affect profit margins, such as price caps, only temporarily, and no barriers to exit. None needs a new spending programme, though removing special levies costs revenue. Better allocation does not mean the state picking which firms should grow, and even well-founded subsidies, such as for research, should be weighed against their unintended effect on allocation. Our evidence is descriptive: it shows that allocation did not improve, not which policy is responsible.

Equal rules, open entry and exit; not the state picking the firms that grow

1 Introduction

A deficit near 7% of GDP: can growth come without new spending?

Hungary runs a large budget deficit. For 2026, the Ministry of Finance projects 7.5% of GDP (Pénzügyminisztérium, 2026) and the central bank 6.9% (Magyar Nemzeti Bank, 2026), well above the 3% reference value of the Maastricht criteria. Bringing the deficit down limits how much the state can spend to support growth. This raises a practical question: are there resources already in the economy that better economic policy could mobilise, without new spending?

This note looks for one such resource. It asks how well the Hungarian economy allocates the workers and capital it already has, whether this has changed over time, and how much output a better allocation could add.

Labour is running out and investment is already high: growth has to come from productivity

Growth can come from more workers, more capital, or higher productivity. More workers is no longer an option: employment rose by almost a fifth between 2010 and 2023, and the working-age population is now shrinking.[1] More capital needs investment that has to be financed, and Hungary already invests heavily. In 2018–2022, the investment of non-financial corporations was 14–16% of GDP a year, compared with 12–13% in the EU and 9–11% in Poland, and government investment added a further 5–6% of GDP. Business investment has fallen since 2022, to 14% of GDP in 2024, but it remains above the EU average.[2] Growth must therefore come mostly from productivity, and a country that invests this much should ask whether its workers and capital end up in the firms that use them best.

Productivity grows when firms improve, or when resources move to better firms

Aggregate productivity rises in two ways. Firms can become more productive by adopting new technology, improving management or finding new markets. Or labour and capital can move from less productive to more productive firms in the same industry, as productive firms grow, weak firms shrink or exit, and new firms enter. This second channel, *reallocation*, requires markets in which good firms can grow and weak firms can shrink or exit.

Neither channel is separate from investment. New investment is itself a decision about which firms grow: capital that goes to productive firms raises aggregate productivity, while capital that goes to weak firms, or to firms favoured for reasons other than productivity, does not. So the question is not only how much Hungary invests, but which firms invest.

Reallocation can also occur between industries, as workers move from manufacturing to services, or from traditional to knowledge-intensive industries. This note leaves that channel aside and compares firms within the same industry. The Hungarian evidence supports this choice: Muraközy et al. (2018) found a clear shift of employment from manufacturing to services until 2010, broadly stable sector shares after, and the vast majority of productivity growth occurring within industries.

Misallocation: rules that keep inputs in weaker firms

When rules, regulations or interventions make it harder for productive firms to

[1] Employed persons aged 20–64: 3.83 million in 2010, 4.55 million in 2023 (Eurostat `lfsi_emp_a`). Population aged 15–64: 6.87 million in 2010, 6.18 million in 2025 (Eurostat `demo_pjanbroad`).

[2] Eurostat, gross fixed capital formation (P51G) as a share of GDP: non-financial corporations (S11, `nasa_10_nf_tr`), general government (S13, `gov_10a_main`), GDP from `nama_10_gdp`. Total investment, including households, was 25–28% of GDP in Hungary and 21–22% in the EU.

expand or for weak firms to shrink, labour and capital remain where they produce less than they could. Economists call this *misallocation*. Over the past decade, Hungarian firms have faced frequent changes in regulation, sector-specific taxes and price caps: on fuel and basic foods in 2021–2023, and on mortgage interest rates from 2022. Each measure had its own goal. Taken together, they may also have changed which firms grow.

Earlier work: reallocation weakened in Hungary after 2013, while gaps narrowed in the region

Hsieh and Klenow (2009) show with firm data that misallocation can lower aggregate productivity substantially. Restuccia and Rogerson (2017) review its causes, including taxes and regulations that depend on firm size or identity, financial frictions and weak property rights.

For Hungary, Muraközy et al. (2018) found strong reallocation before 2008 and much weaker reallocation after 2013, while Koren and Rátfai (2026) show that the employment growth advantage of the most productive Hungarian firms has shrunk steadily since the 1990s. For the region, CompNet data show that the countries that joined the EU in 2004 narrowed the productivity gaps between their firms towards the level of the old member states (Grassi, 2026). We add three things: we extend this evidence to 2023, compare Hungary with Poland and the region on the same measures, and separate foreign-owned from domestic firms. Appendix A reviews the related literature in more detail.

Three exercises on all Hungarian firms, 2004–2023

We use the balance sheets of all Hungarian double-entry firms from 2004 to 2023, together with CompNet data for Poland and the CEE region, and run three exercises, each addressing one question. Exercise 1 asks how large productivity gaps are between firms in the same industry, a first sign of how well resources are allocated, and compares Hungary with Poland and the region. Exercise 2 asks how much productivity growth comes from resources moving between firms, by decomposing growth into firms improving and resources moving. Exercise 3 asks how much is at stake, using the framework of Hsieh and Klenow (2009) to estimate how much more the same inputs would produce if they were where they are most productive.

Data, three exercises, synthesis, policy and caveats; details in the appendix

The rest of the note is organised as follows. Section 2 describes the data. Section 3 defines four terms, presents the three exercises, compares foreign-owned and domestic firms, and applies two of the exercises to four industries. Section 4 brings the results together, Section 5 discusses policy implications and Section 6 sets out the caveats. Appendix A reviews the related literature, Appendix B describes the data sources, variable definitions and sample rules, Appendices C–G give the methods, validation and robustness checks of each exercise, and Appendix H explains how to reproduce the results. The main-text figures also appear in the slide deck that accompanies the note.

2 Data

Balance sheets of all Hungarian double-entry firms, 2004–2023

Our main source is the Hungarian balance-sheet panel (Mérleg LTS, version 5.9.0) for 2004–2023, which covers all firms that keep double-entry books. KRTK compiled it from corporate tax returns up to 2014 and from Opten company

records from 2015 onwards, with contributions from CEU MicroData. For each firm and year, we observe sales, material costs, value added, employment, labour costs, depreciation, tangible and intangible assets, equity, liabilities and the industry code.

Firms with 20+ employees for the comparison with CompNet, 5+ for the market economy

Each exercise uses the sample that suits its method and comparison. Exercise 1 follows the CompNet sample definition: business entities with at least 20 employees in the industries CompNet covers, about 9,500–13,500 firms a year. Exercises 2 and 3 cover the market economy (NACE 05–82 and 95, excluding 53 and 64–66), firms with at least five employees, following Muraközy et al. (2018): 46,000–62,000 firms a year. All three exercises compare firms within the same two-digit industry, so movements of workers and capital between industries are not captured. Appendix B describes the data sources, variable definitions and sample rules in detail.

CompNet: the same measures for Poland and the region, to 2019

For the international comparison we use CompNet, which computes the same firm-level indicators with common code in each participating country (CompNet, 2021). We use its 8th vintage, restricted to firms with 20 or more employees, as aggregated by Basile Grassi for Grassi (2026). The Hungarian series start in 2004 and the Polish in 2002; all end in 2019 (2018 for most old member states).

We compare Hungary with Poland and with the average of the Central and Eastern European countries that joined the EU in 2004 and report to CompNet (the *CEE average*), using the average of the old member states as a reference level. Until 2014, CompNet's Hungarian data come from the same tax returns as ours, so we check our Hungarian series against CompNet's for 2004–2019 (Appendix C); 2015–2019, after our source switch, is the more independent part of that check.

Prices matter only for growth over time; 2023 data are provisional

Exercises 1 and 3 compare firms within the same industry and year, so a price index common to all firms in an industry and year does not affect their results. Differences in firm-specific prices and markups, however, remain: we observe revenue rather than physical output, so a firm that charges higher prices looks more productive. Exercise 2 measures growth over time and therefore needs value added at constant prices. We use two price indices, industry value-added deflators from the national accounts and industry producer price indices, and report results for both, because they diverge in the high-inflation years 2021–2023. Value added in our sample grows at the same rate as in the national accounts until 2022, but more slowly in 2023, so we treat the 2023 results as provisional.

3 Three exercises

This section presents the evidence. We first define four terms used throughout the note, then take the three exercises in turn. For each exercise we explain what we compute and why it is informative about allocation, describe the sample, present the results and discuss what they mean. All three exercises address the same question: are workers and capital used where they are most productive? Wider

productivity gaps between firms in the same industry, a smaller contribution of reallocation to productivity growth, and a larger efficiency gap would each point to less efficient allocation. Section 3.5 compares foreign-owned and domestic firms, and Section 3.6 applies Exercises 2 and 3 to four industries. Methods, validation and robustness checks are in Appendices C–G.

3.1 Four terms

The following four terms are used throughout the note.

Productivity of a firm. The value added a firm produces from its inputs. *Labour productivity* is value added per employee. *Total factor productivity* (TFP) also takes capital into account: it is value added relative to a combination of labour and capital, each weighted by its share in costs. We measure TFP as a *Solow residual*, the part of value added not explained by labour and capital. Because we observe revenue rather than physical output, this is *revenue productivity*: a firm that charges higher prices looks more productive.

Productivity gaps within an industry. In a market that works well, productive firms expand and weak firms shrink or exit. Because we measure revenue productivity, in a market without frictions it would be nearly equal across firms in the same industry: a more productive firm expands and lowers its price until its revenue per unit of input matches that of its competitors. A wide, persistent gap therefore suggests that something prevents resources from flowing to where they are most productive, although measurement error and differences in market power can also widen it. We measure the gap as the spread of TFP across firms in the same industry and year, relative to its average.

Reallocation. Aggregate productivity in an industry is the employment-weighted average of its firms' productivity. It rises when firms become more productive, when employment shifts towards more productive firms among continuing firms, or when weak firms leave the sample and new ones enter (entry and exit). The last two, reallocation among continuing firms and entry and exit, together make up the part of productivity growth that comes from resources moving between firms. We call this reallocation in the broad sense; where we mean the shift among continuing firms only, we say so.

Misallocation and the efficiency gap. If every firm in an industry earned the same return on its last worker and its last forint of capital, no reallocation of inputs could raise output. The *efficiency gap* measures how much more an industry would produce at that point with the same inputs, as a share of its actual value added: a potential gain, not a loss. It is a model benchmark: differences in returns can also arise from adjustment costs, risk and measurement error. For this reason, its change over time is more informative than its level.

3.2 Exercise 1: how far apart are firms in the same industry?

Wider productivity gaps within an industry signal that resources may not flow to the most productive firms

The first exercise asks how large productivity gaps are between firms in the same industry. If workers and capital move freely, productive firms expand until the return on additional inputs falls to that of their competitors, and the gaps between firms in an industry narrow over time. A gap that remains wide suggests that resources are not moving to where they are most productive (Hsieh and

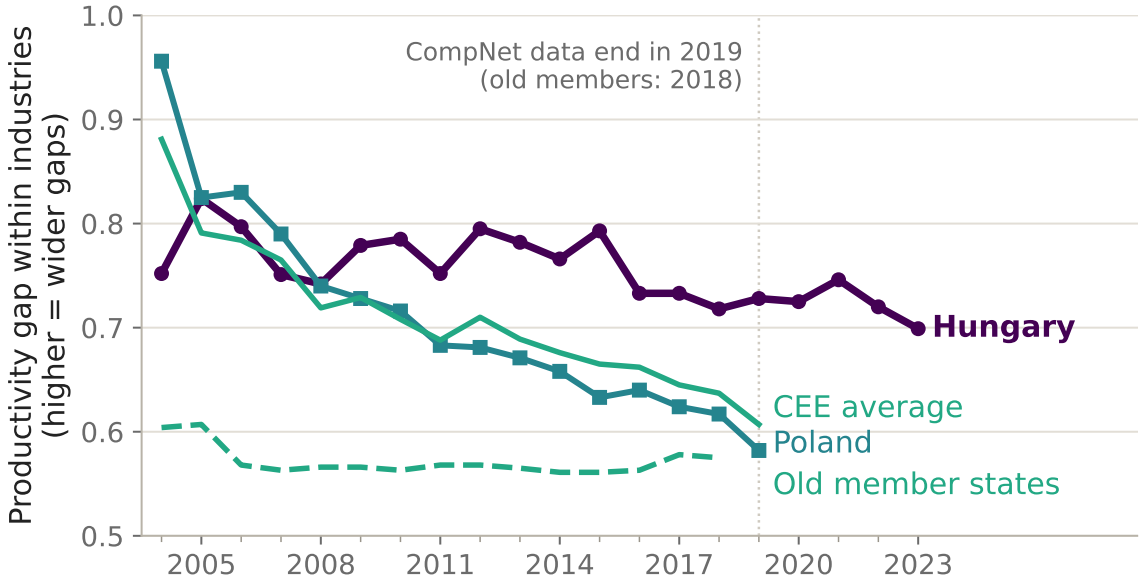


Figure 1: Exercise 1: productivity gaps between firms in the same industry, 2004–2023

Notes: Standard deviation divided by the mean of firm-level TFP within two-digit industries, aggregated across industries as a root mean square weighted by the number of firms. TFP is a Solow residual with time-varying cost-share weights, computed from industry-year medians of cost shares, with the cost of capital equal to depreciation. Firms with 20 or more employees in CompNet industries. Hungary: own calculation from the Mérleg LTS panel. Poland, CEE average and old member states: CompNet 8th vintage, aggregated by B. Grassi; CompNet's cost of capital also includes interest paid and an imputed interest on equity (CompNet, 2021). The CEE average covers the 2004 EU joiners reporting to CompNet and includes Hungary. The old member states are the nine EU-15 countries with data, shown for years in which at least five report. Figure 6 shows a version in which the cost of capital also includes an imputed financing cost.

Klenow, 2009). For each firm, we compute total factor productivity as a Solow residual, weighting labour and capital by their shares in costs in the industry and year. Within each two-digit industry and year, we divide the standard deviation of firm TFP by its mean; we call this the spread of TFP. A spread of 0.7 means that the typical deviation from the industry average is 70% of that average. We aggregate across industries, weighting each by its number of firms. This is the measure CompNet uses, so Hungary can be compared directly with Poland and the region. For Hungary we use all firms with 20 or more employees in the industries CompNet covers, for 2004–2023. For Poland, the CEE average and the old member states we use CompNet's series, to 2019 (CompNet, 2021; Grassi, 2026). These follow the same definitions, except that CompNet's cost of capital also includes interest paid and an imputed interest on equity, while our baseline uses depreciation only. Appendix C describes the method and checks our Hungarian series against CompNet's own.

Hungary's gaps stayed wide (0.72–0.75) and by 2019 were clearly wider than Poland's

Figure 1 shows the spread of TFP between firms in the same industry, aggregated across industries; higher values mean wider gaps. In the mid-2000s, Hungary, Poland and the CEE average were at roughly the same level, 0.76–0.80, well above the old member states at 0.58. On this measure, Poland's spread then fell almost every year, and by 2018–2019 it had reached the level of the old member states. The CEE average fell too, though less. Hungary's spread rose during the 2008–2009 crisis, fell back in 2016, the year after our data source changed, and has stayed at around 0.72–0.75 since. CompNet's own Hungarian series falls in the same year, so the 2016 drop is unlikely to be an artefact of the switch.

By 2019 the Hungarian gap was clearly larger than the Polish one, and CompNet's own Hungarian series gives the same picture. On our series the 2016–2019 average was 0.73, against 0.62 for Poland. Because the CEE average includes Hungary, the comparison with the region understates the difference.[3] Adding an imputed financing cost to the cost of capital raises the Hungarian line somewhat in years of high interest rates (2009–2013 and 2022–2023) but leaves the rest of the picture unchanged (Appendix C).

Consistent with resources held in weaker firms, but other explanations remain possible

Persistent gaps mean that more productive Hungarian firms did not grow enough to close them, that less productive ones did not shrink enough, or both. This is consistent with an economic environment that kept resources in less productive firms. It is also consistent with other explanations, such as differences in the composition of activities within industries, in prices and markups, or in measurement. Exercise 1 cannot distinguish between them.

3.3 Exercise 2: how much growth comes from resources moving?

Productivity grows through two channels: firms improving and resources moving

Exercise 1 describes the state of allocation in a given year. Exercise 2 looks at the flow: how much productivity growth comes from workers moving between firms and from firms entering and exiting. An industry's productivity can rise even if no firm becomes more productive: when employment shifts from weaker to stronger firms, when weak firms exit, or when productive firms enter. If this contribution shrinks, one of the two channels of productivity growth weakens. We first decompose productivity growth into its components, then track the contribution of resources moving over two decades, and finally ask how much of its change comes from foreign-owned firms.

Four components: firms improving, employment shifting, entry and exit

Following Foster et al. (2001), we decompose the change in an industry's productivity over three years into four components: (i) *within firms*, as continuing firms improve; (ii) *reallocation among continuing firms*, as employment shifts between them towards more productive ones (the between term, plus a cross term for firms whose productivity and employment share change together); (iii) *entry*; and (iv) *exit*. We decompose within each two-digit industry and aggregate across industries with employment weights, and report annualised values. Components (ii)–(iv) together make up the growth that comes from resources moving: reallocation in the broad sense. The entry and exit components measure how far entrants and exiters differ from the industry average, weighted by their size, not how many firms enter or exit: frequent entry and exit of average firms would add nothing. Because the decomposition uses employment weights, it shows where workers go; capital enters through the productivity measure, TFP, and through the entry and exit of firms. Appendix E gives the formulas.

Market-economy firms with 5+ employees, 2004–2022, three-year windows

We use all market-economy firms with at least five employees, 2004–2022, in three-year windows. Results for 2022–2023 are reported in Table 1 but treated as provisional. Value added is at constant prices, deflated with industry value-

[3] The difference in levels holds for all three productivity measures that CompNet publishes: labour productivity, TFP with fixed weights and TFP with cost-share weights. The contrast in trends, with all three countries starting at a similar level and Poland's gaps narrowing much more, appears only on the cost-share measure. On none of the three measures did the Hungarian gaps narrow after 2019 (Appendix D).

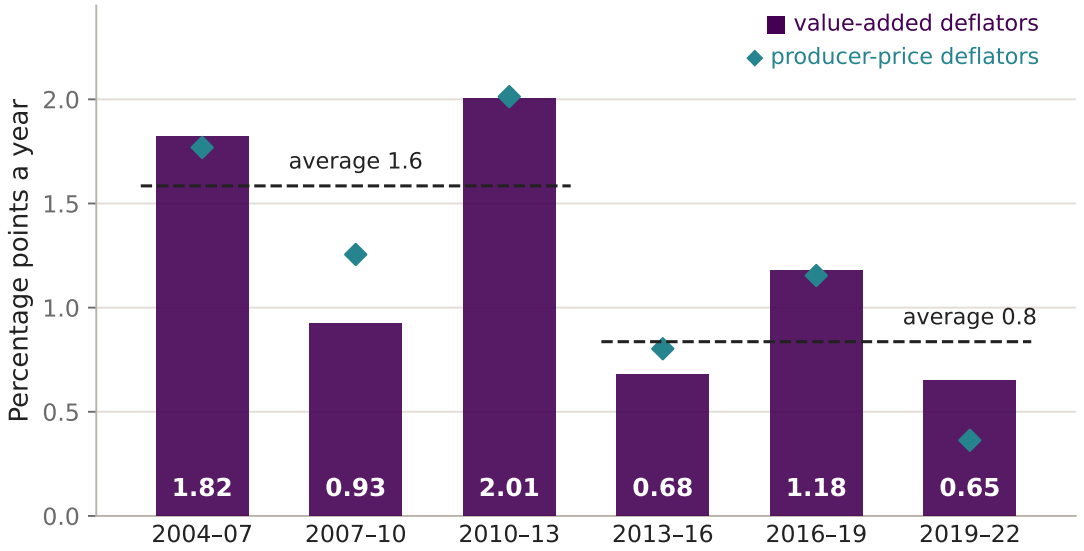


Figure 2: Exercise 2: contribution of resources moving between firms to productivity growth, by three-year window

Notes: Reallocation among continuing firms (between and cross terms) plus entry and exit, in percentage points a year. Foster–Haltiwanger–Krizan decomposition within two-digit industries, with employment weights; market economy, firms with 5+ employees. Bars: value-added deflators; diamonds: producer price indices. Dashed lines: averages for 2004–2013 and 2013–2022, with value-added deflators. The provisional one-year window 2022–2023 is not shown. Period averages of all components are in Table 1.

added price indices from the national accounts (Eurostat, 2026); Appendix F uses producer price indices instead. Total growth is the change in the employment-weighted average of firms' log TFP within industries. This differs from aggregate TFP in the national accounts, which weights firms by output and covers the whole economy, and it can grow much faster, as in 2016–2019. We use it only to decompose growth into its sources. Entry and exit refer to firms entering or leaving the observed sample of an industry. Most are firms appearing in or disappearing from the data. Some, however, are continuing firms that changed industry: up to two-fifths of entrants and a third of exiters in a window (Appendix E). And some exits are firms that stopped reporting or fell below the size threshold, rather than closures.

The fall after 2013 holds in every alternative we tried

These choices affect the levels of the components, but not the main finding. Appendix F shows the results under alternative choices: producer price indices instead of value-added deflators; cost-share weights that exclude the financing cost of capital; entry and exit defined by crossing the five-employee threshold; excluding firms that change industry; splitting the cross term equally between firms improving and reallocation; and the decomposition of Melitz and Polanec (2015). In every case, the contribution of resources moving is lower in 2013–2022 than in 2004–2013. Across the specifications it falls by 0.7–1.0 points a year. Excluding firms that change industry raises the earlier contribution and leaves the later one almost unchanged, while the Melitz–Polanec decomposition makes the decline larger: there the contribution turns slightly negative after 2013.

Growth from resources moving almost halved after 2013: from 1.6 to 0.8 points a year

Figure 2 shows how many percentage points a year resources moving added to productivity growth in each three-year window. The series is volatile: 1.82 points a year in 2004–2007, 0.93 in 2007–2010, 2.01 in 2010–2013, 0.68 in

Table 1: Exercise 2: sources of productivity growth, percentage points a year

	2004–13	2013–22	2013–16	2016–19	2019–22	2013–23
Firms improving (within)	0.55	2.84	1.85	3.95	2.71	1.76
Between	1.12	0.84	0.65	0.80	1.06	0.85
Cross	−0.53	−0.60	−0.44	−0.41	−0.95	−0.68
Reallocation among continuing firms (between + cross)	0.59	0.23	0.21	0.39	0.10	0.17
Entry	−0.31	0.00	−0.34	0.44	−0.10	−0.10
Exit	1.31	0.60	0.81	0.35	0.65	0.60
Entry and exit	1.00	0.60	0.47	0.79	0.55	0.50
Resources moving (reallocation + entry and exit)	1.58	0.84	0.68	1.18	0.65	0.67
Total	2.14	3.67	2.53	5.13	3.36	2.43
Share of total growth from resources moving	74%	23%	27%	23%	19%	28%
Resources moving, producer prices	1.68	0.77	0.80	1.15	0.36	0.66
Share of total growth, producer prices	72%	26%	24%	23%	102%	26%

Notes: Foster–Haltiwanger–Krizan decomposition within two-digit industries, aggregated with industry employment weights; annualised log changes ×100. Three-year windows: 2004–13 and 2013–22 are averages of three windows each; 2013–16, 2016–19 and 2019–22 are single windows; 2013–23 adds the provisional one-year window 2022–23, weighted by window length. Value-added deflators unless otherwise stated. Shares of total growth are not meaningful when total growth is close to zero (producer prices, 2019–22: total 0.35). Market-economy firms with at least 5 employees in at least one end-year of the window; Appendix F defines entry and exit by crossing the 5-employee threshold instead.

2013–2016, 1.18 in 2016–2019 and 0.65 in 2019–2022. Part of this volatility reflects the business cycle and macroeconomic shocks, such as the 2008–2009 crisis and the years 2020–2022, so a single window says little. We therefore compare averages over longer periods. Averaged before and after 2013, the contribution fell by almost half, from 1.6 points a year in 2004–2013 to 0.8 in 2013–2022 (1.7 and 0.8 with producer price indices). Every window after 2013 is below the earlier average. The two weakest are 2013–2016 and 2019–2022; the second covers the pandemic, the energy shock and the price caps.

Table 1 shows all components before and after 2013, and for each later window. Productivity growth itself did not collapse: after 2013 most of it came from firms improving themselves, 2.8 points a year in 2013–2022, against 0.6 before 2013. This contrast mostly reflects the crisis: within-firm growth was −2.8 points a year in 2007–2010, compared with 3.2 in 2004–2007 and 1.2 in 2010–2013. The same window pulls down total growth before 2013, so the share of growth coming from resources moving (74% before 2013, 23% after) overstates the change, and we do not rely on it. The fall in the contribution of resources moving, however, does not depend on the crisis window: excluding 2007–2010, it is 1.9 points a year before 2013, compared with 0.8 after.

Weak firms exiting and workers moving to better firms both contributed less after 2013

Three details help to interpret the numbers. First, the entry term is negative in four of the six windows: new firms are, on average, less productive than the incumbents of their industry when they start. Exiting firms are also less productive than average, so their exit raises productivity. The gain from entry and exit therefore comes from exit. The exception is 2016–2019, when entry contributed more (0.44 points) than exit (0.35).

Second, between 2004–2013 and 2013–2022 the contribution of resources moving fell by about 0.75 points a year. About half of the fall comes from entry and exit: the exit of weak firms contributed much less (1.3 points a year before 2013, 0.6 after), only partly offset by entrants pulling productivity down less. The other half comes from reallocation among continuing firms, mainly

the between term, which captures employment shifting towards initially more productive firms (1.1 points a year before 2013, 0.8 after). In other words, employment shifted less towards the firms that were already more productive.

Third, the cross term, which is negative when firms whose TFP rises lose employment share, is about the same on average before and after 2013, but it is most negative in 2019–2022 (–0.95). A negative cross term can mean that workers do not follow productivity gains, but it can also reflect adjustment lags, labour hoarding during a crisis, or measurement error, because employment enters both productivity and the employment shares. If the cross term is split equally between firms improving and reallocation, the fall in the contribution of resources moving is about the same size.

This pattern is consistent with Koren and Rátfai (2026), who find that the employment growth advantage of the most productive Hungarian firms has shrunk.[4] Whether the weakness of 2019–2022 persists once the crisis years have passed is an open question: the weak window of 2013–2016 was followed by a partial recovery. Section 3.5 breaks the fall down by ownership.

3.4 Exercise 3: how much output could better allocation add?

How much more the same inputs would produce if returns were equal across firms

The last exercise asks how much is at stake. We follow Hsieh and Klenow (2009). Suppose labour and capital were reallocated within an industry until every worker and every forint of capital earned the same marginal return in every firm. The calculation asks how much more the industry would then produce with the same total labour and capital. In the model, each firm faces its own effective tax or subsidy, on its output or on its capital. In practice this can be a tax rate, an exemption, a regulation, a price cap, cheaper credit or a payment the firm must make to operate. Firms facing high effective taxes earn high returns and are too small; favoured firms earn low returns and are too large. What lowers aggregate output is the dispersion of these effective rates across firms, not their average level. The benchmark is therefore equal treatment, the same effective rate for every firm, not a lower rate for firms that look productive. The model assumes that firms within an industry sell differentiated products, with an elasticity of substitution of 3 as in Hsieh and Klenow, and it aggregates industries using their value-added shares. We treat the level of the gap as a model benchmark and focus on how it changed over time.

Market-economy firms with 5+ employees, 2004–2023; manufacturing shown separately

We use all market-economy firms with at least five employees, 2004–2023, and show manufacturing separately. Following Hsieh and Klenow (2009), we trim firms with extreme measured productivity before the calculation, 1% at each tail of revenue and physical productivity. The trim removes about 3% of firms but 31–44% of value added, depending on the year, because it includes many of

[4] Exercises 1 and 3 measure how unequal productivity is across firms in a given year. Koren and Rátfai (2026) look at change instead: they ask whether firms with higher value added per worker than others in their industry grow faster over the following three years, and find that this growth advantage has shrunk. The two are sides of the same mechanism. In a market that works well, firms with high revenue productivity grow, and as they grow their revenue per unit of input falls back towards that of their competitors, so the gaps narrow. When productive firms no longer outgrow the rest, the gaps that new shocks create remain open. The between term of Exercise 2, employment shifting towards initially more productive firms, is the closest aggregate counterpart of their growth advantage, and it too fell after 2013.

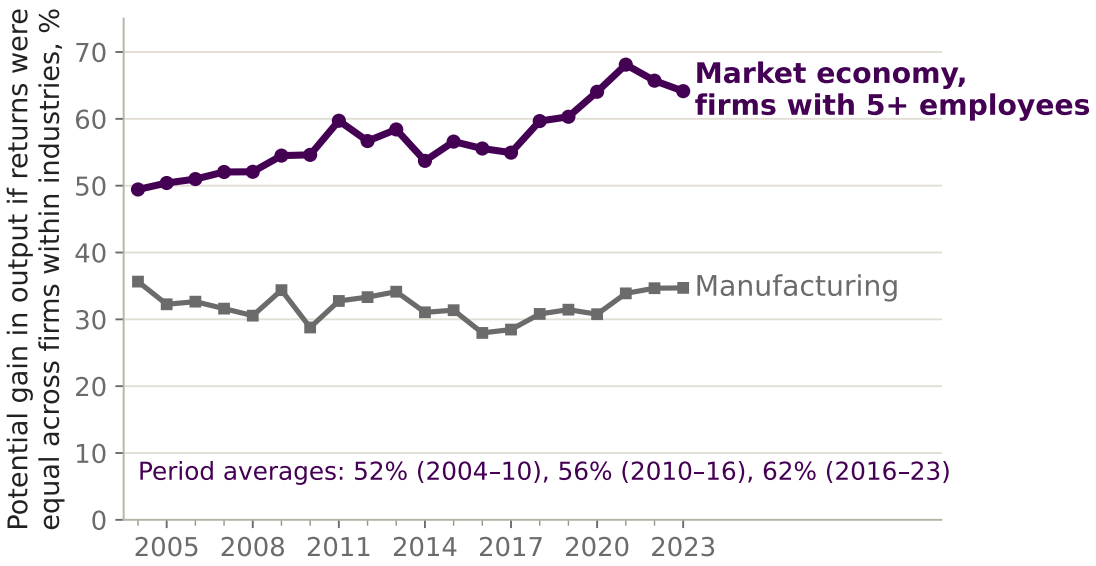


Figure 3: Exercise 3: efficiency gap, the potential gain in output from equalising returns within industries

Notes: Percentage increase in value added if marginal returns to labour and capital were equalised across firms within each two-digit industry (Hsieh and Klenow, 2009). Elasticity of substitution $\sigma = 3$; industries weighted by value-added shares; 1% of each tail of revenue and physical productivity trimmed. Market economy and manufacturing; firms with five or more employees. Period averages in the figure: 2004–2010, 2010–2016 and 2016–2023, which share their boundary years. 2023 is provisional. Results by sector are in Figure 10.

the largest firms. The 2023 data are provisional; they matter less here than in Exercise 2, because the model compares firms within the same year. Appendix G gives the formulas and modelling choices, and shows how the results depend on trimming and on the elasticity of substitution.

The efficiency gap rose by 6–10 percentage points of value added; in manufacturing it was roughly unchanged

Figure 3 shows the efficiency gap. In the market economy (purple line) it was 52% of value added in 2004–2010. It rose by about six points by 2016–2019, and by about ten points when 2020–2023 is included. In manufacturing (grey line) it was roughly unchanged over the period as a whole. The rise is largest in construction and utilities, and also visible in information and communication, accommodation and food services, professional services, real estate and wholesale and motor trade (retail hardly changed). It reflects rising gaps within these sectors rather than shifts in their weights (Figure 10). Among firms with 20 or more employees in the industries CompNet covers, which exclude utilities, the gap rose by four points, and in manufacturing not at all (Table 8). The efficiency gap therefore grew outside manufacturing, and less among firms with 20 or more employees; part of this difference comes from the exclusion of utilities, so we do not read it as a pure size effect.

Capital became more unevenly allocated across firms; labour did not

A more direct approach looks at the two inputs separately. For each firm, we compute value added per unit of capital and per unit of wage bill, the returns on the two inputs, and measure how unequal these returns are across firms in the same industry. This needs no choice of weights, because the factor shares are the same for all firms in an industry and drop out. Between 2004–2010 and 2010–2016 the spread of returns on capital rose from 1.45 to 1.58 (standard deviation of logs) and then stayed at that level (1.57 in 2016–2023). The spread of returns on labour did not rise (0.55, 0.53 and 0.53). Figure 4 shows the yearly series: the spread of returns on capital rose steadily from 2004 to 2013 and

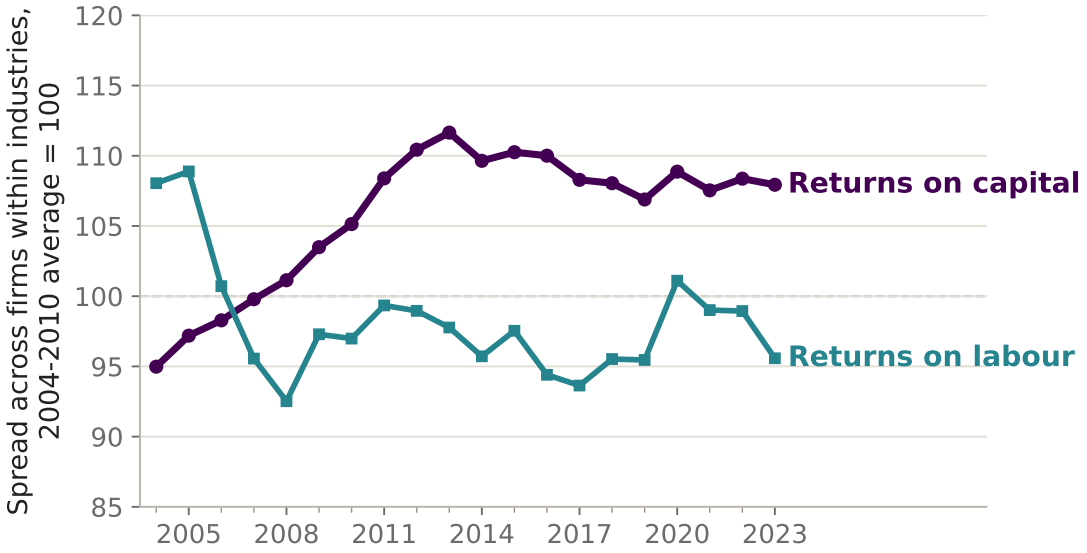


Figure 4: Exercise 3: spread of returns on capital and on labour across firms, 2004–2023

Notes: Standard deviation across firms of log returns on capital (value added divided by capital) and on labour (value added divided by the wage bill), measured as deviations from industry-year means, as Hsieh and Klenow (2009, Table II) do for revenue productivity. Market economy, firms with five or more employees; same sample as Figure 3. Each series is indexed to its own 2004–2010 average (=100; 1.45 for capital, 0.55 for labour), because the levels of the two series are not comparable.

remained high, so the rise began before 2010. This is the pattern Gopinath et al. (2017) found in Spanish manufacturing, with similar trends in Italy and Portugal, and Gamberoni et al. (2016) in five euro-area countries: capital became more unevenly allocated, while labour did not. The levels of the two spreads cannot be compared, because capital is costly to adjust, so returns on it are more dispersed even without distortions (Asker et al., 2014). Only the change is informative, and part of it may reflect more volatile conditions rather than policy.

The timing differs from that of the efficiency gap. The spread of returns on capital stopped rising around 2013, and the spread of revenue productivity hardly changed (Appendix G), yet the efficiency gap kept rising after 2016. The later rise therefore comes mainly from how the effective taxes line up with firms' underlying productivity, not from wider spreads of returns. This is the part of the model most sensitive to the tails of the distribution and to trimming, and the model itself does not attribute the efficiency gap to either input.

Exercise 3 translates the productivity gaps of Exercise 1 into output, for a wider set of firms

Exercises 1 and 3 start from the same object: how unequal revenue productivity is across firms in an industry. Exercise 1 measures this spread and compares it across countries. Exercise 3 puts it into a model that translates it into the output better allocation could add. The two differ in four ways. First, the samples: Exercise 1 covers firms with 20 or more employees in CompNet's industries and weights industries by their number of firms; Exercise 3 covers market-economy firms with five or more employees and weights industries by value added. Second, the statistic: Exercise 1 uses the spread of TFP levels relative to the mean, Exercise 3 the full distribution of log revenue productivity and of firms' underlying productivity. Third, the cost of capital: depreciation only in the baseline of Exercise 1, depreciation plus an imputed financing cost in Exercise 3. Fourth, Exercise 3 has no comparison with Poland.

For Hungary, the two agree where they overlap. Among firms with 20 or more employees in CompNet's industries, the sample of Exercise 1, the efficiency gap rose by only four points, and in the main sample the spread of revenue productivity itself hardly changed (Appendix G). This matches Exercise 1, where the Hungarian gap stayed wide but did not widen. The larger rise in Exercise 3 comes from smaller firms and industries outside manufacturing, and from how the effective taxes relate to firms' underlying productivity. Hence Exercise 1 shows that the gaps did not close, and Exercise 3 shows what these gaps cost and that the cost has risen.

Back to the 2004–2010 allocation: output about 1–2% of GDP higher, a one-off rise that persists

A useful benchmark is Hungary's own past. If the efficiency gap returned to its 2004–2010 average, the same workers and capital would produce about 1–2% of GDP more. We call this the *reserve*. It is a one-off rise in the level of output that then persists, not additional growth every year. If it took five years to reach, it would add 0.2–0.4 percentage points to annual growth over those five years.

The step from the efficiency gap to GDP is as follows. With 2016–2023 as the later period, the gap of 62% against 52% in 2004–2010 means that output would be 6.3% higher (1.616/1.520 – 1) for the firms in the calculation. These firms produce two-thirds of the value added of all market-economy firms with five or more employees, and that value added is about 40% of GDP. Applied to the firms kept, the reserve is 1.7% of 2023 GDP; applied to all firms with five or more employees, which assumes that the trimmed firms would gain in the same proportion, it is 2.5%. With the pre-pandemic years 2016–2019 as the later period, it is 1.0–1.5% of GDP. The band of 1–2% covers the pre-pandemic comparison and the lower part of the comparison with 2016–2023. We put less weight on the upper part, because 2020–2023 includes the crisis years and the provisional 2023 data, which also set the value-added base.

The figure is sensitive to the model's choices and to the years compared, so we treat it as an order of magnitude rather than a precise estimate. Across the variants in Appendix G, the reserve ranges from 0.8% to 5.1% of GDP with 2016–2023 as the later period, and from 0.2% to 2.0% with 2016–2019. With an elasticity of substitution of 5 instead of 3, the gap is larger, and its rise and the reserve are about twice as large (5.1% of GDP). Without trimming, the pre-pandemic comparison almost disappears (about 0.25% of GDP). The direction of the change is robust; its size, and hence the reserve, is not.

Poland's efficiency gap rose too: read Exercise 3 as an order of magnitude

Using the same method, Gradzewicz (2026) finds that the efficiency gap in Poland also rose, from close to 40% of value added in the 1990s to 70–75% in the early 2020s. This measure therefore does not show the contrast between Hungary and Poland found in Exercise 1, and the rise over time cannot be read as specific to Hungary. The two Polish results, narrowing productivity gaps and a rising efficiency gap, are based on different samples and definitions, and we cannot reconcile them. The levels of the two studies are not comparable either. This is one more reason to read Exercise 3 as an order of magnitude.

3.5 Foreign-owned and domestic firms

Weak allocation is not only a problem of domestic firms

Foreign-owned firms employ 37–39% of the workers in our sample of firms with five or more employees, depending on the window. Is weak allocation a problem of domestic firms, which are often small, or does it also affect foreign-owned firms? Table 2 breaks down both the contribution of resources moving (Exercise 2) and the efficiency gap (Exercise 3) by ownership.

Table 2: Foreign-owned and domestic firms: resources moving (Exercise 2) and the efficiency gap (Exercise 3)

A. Exercise 2: contribution of resources moving, percentage points a year	2004–13	2013–22	Change
All firms	1.58	0.84	−0.75
Between domestic and foreign-owned firms	0.20	0.00	−0.20
Within domestic firms	0.94	0.69	−0.26
Within foreign-owned firms	0.44	0.16	−0.29
Domestic firms only, separate decomposition	1.50	1.12	−0.38
Employment share of foreign-owned firms, %	37	39	2
B. Exercise 3: efficiency gap, % of value added	2004–10	2016–19	2016–23
All firms	52	58	61
Domestic firms	49	58	61

Notes: Panel A: reallocation plus entry and exit as in Table 1, averages of three three-year windows. The three margins add up to the value for all firms. Between domestic and foreign-owned firms: within each industry, the gain from the two groups' changing employment shares and average productivity. Within each group: reallocation, entry and exit among the group's own firms, weighted by the group's employment share at the start of the window. Domestic firms only: the decomposition redone without foreign-owned firms. Employment share of foreign-owned firms at the start of each window. Ownership (ultimate owner) is measured at the start of each window, for entrants at its end. The ownership data change source in 2015 and again after 2020: the foreign share of value added falls from 68% to 60% across 2015–2016 and the foreign share of firms from 9.1% to 7.8% across 2020–2021. The foreign contribution is lower in every window after 2013, not only after these breaks, but we read the split as indicative. Panel B: efficiency gap as in Figure 3, period averages, firms with 5+ employees; both rows cover the industries of the domestic sample, weighted by their value-added shares in the full sample, so the values for all firms differ slightly from Figure 3.

Foreign-owned firms stopped gaining ground on domestic firms, and resources moved less among them

The contribution of resources moving splits exactly into three margins (Panel A). The first is between the two groups: in an industry where foreign-owned firms are more productive on average, the economy gains when they employ a larger share of its workers. The second and third are within each group: workers moving between domestic firms, and between foreign-owned firms, including entry and exit. Before 2013 the margin between the groups added 0.2 points a year, as foreign-owned firms gained employment share in every window; after 2013 it added nothing. Reallocation within foreign-owned firms fell from 0.44 to 0.16 points a year, by about two-thirds, and within domestic firms from 0.94 to 0.69, by about a quarter. Of the fall of 0.75 points, 0.20 comes from the margin between the groups, 0.29 from within foreign-owned firms and 0.26 from within domestic firms. Foreign-owned firms thus account for about two-thirds of the fall (0.49 of 0.75), well above their share of employment, in two ways: they stopped expanding at the expense of less productive domestic firms (the margin between the groups, 0.20), and resources moved less among them (0.29, or about 40% of the fall on its own). Counting the margin between the groups as a foreign-owned margin is a reading: it is foreign-owned firms that gained employment share in every window before 2013, but the margin is a property of both groups. It is also small and varies across windows (0.15, 0.28 and 0.17 before 2013; −0.06, −0.08 and 0.13 after), so we interpret it as a margin that closed rather than as a large loss. A separate decomposition on domestic firms only gives the same picture: the contribution fell from 1.5 to 1.1 points a year,

by about a quarter.

Among domestic firms the efficiency gap rose at least as much as overall

Panel B compares the efficiency gap. With the same industries and industry weights, the efficiency gap among domestic firms rose from 49% of their value added in 2004–2010 to 58% in 2016–2019 and 61% in 2016–2023, compared with 52%, 58% and 61% for all firms. Domestic firms start from a slightly smaller gap and end at the same level, so the rise among them is at least as large as overall. With the domestic firms' own industry weights the rise looks larger (49% to 64%), but this reflects which industries domestic value added is concentrated in, not a difference within industries.

Foreign-owned firms were on both sides of firm-specific measures

So allocation weakened in both groups, in different ways. Among foreign-owned firms the flow of resources fell by about two-thirds, and Koren and Rátfai (2026) find that their employment growth advantage over non-exporting domestic firms has disappeared. The end of the expansion of foreign-owned firms, which Muraközy et al. (2018) describe before the crisis, appears in our decomposition as the margin between the groups falling to zero. Part of this may come from policy, because foreign-owned firms were on both sides of firm-specific measures. The sector-specific special taxes introduced from 2010 onwards, in retail, telecommunications, energy and banking among others, fell mainly on large, foreign-owned firms. Varju and Papp (2022) argue that their design allowed discrimination against foreign-owned taxpayers; court rulings on this are mixed.[5] At the same time, since 2012 the government has signed strategic cooperation agreements with more than 100 large companies by the end of 2025, nine in ten of them foreign-owned (HIPA Hungarian Investment Promotion Agency, 2025). Both kinds of measure make how a firm is treated depend on who it is rather than on general rules. Among domestic firms the flow slowed less, but the efficiency gap widened at least as much as overall. Our data do not show which measures, if any, caused these changes.

For policy, this suggests that equal treatment matters for foreign-owned firms as much as for domestic ones. Special taxes that single out some firms, and subsidies and individual agreements that favour others, both make effective tax rates differ across firms in the same industry. Measures that burden foreign-owned firms more than their domestic competitors may also have contributed to closing the margin between the groups after 2013: the growth of more productive foreign-owned firms at the expense of less productive domestic ones. Publishing the terms of individual agreements and subsidy decisions would allow their effect on allocation to be assessed.

To sum up the split: the slowdown of the flow came mostly from foreign-owned firms, while the rise in the efficiency gap is at least as large among domestic firms.

[5] The Court of Justice of the EU found the aggregation rule of the 2010 retail tax indirectly discriminatory (Hervis, C-385/12, 2014), but found the progressive telecommunications and retail taxes compatible with EU law (Vodafone, C-75/18; Tesco, C-323/18, 2020). In 2026 the European Commission referred Hungary to the Court over the re-introduced retail tax.

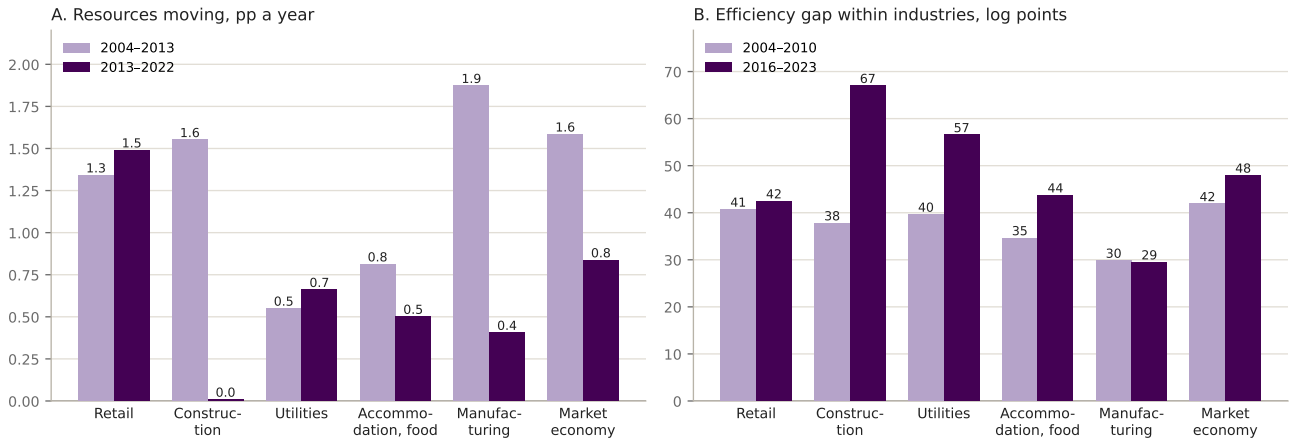


Figure 5: Resources moving and the efficiency gap in four industries

Notes: Panel A: reallocation plus entry and exit (Exercise 2, main specification), in percentage points a year; two-digit industries weighted by their employment, averages of the three-year windows before and after 2013. Panel B: efficiency gap, $100 \ln(TFP^{\text{eff}}/TFP)$ within two-digit industries (Exercise 3, main specification), in log points, weighted by value added, period averages. Log points are smaller than the percentages of Figure 3 (52% is 42 log points), so the two are not directly comparable. Industries (NACE Rev. 2): retail, 47; construction, 41–43; utilities, 35–39; accommodation and food services, 55–56; manufacturing (for comparison), 10–33. Firms with five or more employees.

3.6 Four industries

Policy makers want to know where the problem lies. Figure 5 shows Exercises 2 and 3 for four industries that have been subject to heavy sector-specific intervention since 2010, such as special taxes, regulated prices, price caps and public procurement: retail, construction, utilities, and accommodation and food services. We chose them before computing the results. Manufacturing and the market economy are shown for comparison.

The efficiency gap rose most in construction and utilities, and hardly at all in retail and manufacturing

Figure 5 compares the earlier period (light bars) with the later one (dark bars). Panel A shows the contribution of resources moving to productivity growth (higher is better); Panel B shows the efficiency gap (higher is worse). The efficiency gap rose most in construction (38 to 67 log points) and utilities (40 to 57), less in accommodation and food services (35 to 44), and hardly at all in retail (41 to 42) and manufacturing (30 to 29). The contribution of resources moving is noisier at the industry level, because individual three-year windows vary widely, sometimes switching from positive to negative. Two patterns are clear. In construction the contribution fell from 1.6 points a year to about zero, and in manufacturing from 1.9 to 0.4. In retail it did not fall (1.3 and 1.5).[6] In utilities and in accommodation and food services the windows vary too much to identify a trend.

Construction: both measures got worse

Construction is the industry in which both measures clearly deteriorated. Manufacturing shows the opposite combination: resources moved much less after 2013, but the efficiency gap, which is measured within each year, did not widen.

[6] Muraközy et al. (2018, Chapter 8) find weaker reallocation in retail after 2010 than before the crisis. Our earlier period includes 2010–2013, so the two results are consistent: reallocation in retail weakened around the crisis but did not decline further after 2013. Section 5 returns to retail.

Table 3: What each exercise measures, what it finds, and what it does not tell us

	What it measures	What it finds	What it does not tell us
Exercise 1: productivity gaps	How much productivity differs between firms in the same industry, compared with Poland and the region (firms with 20+ employees; Poland and the region to 2019)	In 2016–2019 Hungary's gaps were about 0.1 wider than Poland's on every measure, and they did not narrow after 2019	Whether the gaps reflect policy, prices and markups, or measurement
Exercise 2: reallocation	How much productivity growth comes from workers moving between firms and from firms entering and exiting (firms with 5+ employees)	1.6 points a year in 2004–2013, 0.8 in 2013–2022; about two-thirds of the fall from foreign-owned firms	Why workers moved less, or how capital moved
Exercise 3: what is at stake	How much more the same inputs would produce, according to a standard model, if returns were equal across firms, and how this changed	The efficiency gap rose by 6–10 points of value added, almost entirely outside manufacturing; back to the 2004–2010 allocation: about 1–2% of GDP	The effect of any specific policy change

4 Putting it together

This section brings the three exercises together: what each one shows and how the findings fit with one another.

One finding from each exercise

Exercise 1: gaps between firms stayed wide. By the late 2010s, productivity gaps within industries were wider in Hungary than in Poland and the CEE region on every measure, while, on our main measure, Poland's were close to the level of the old member states. In Hungary the gaps did not narrow in 2020–2023.

Exercise 2: resources moved less. Reallocation among continuing firms, together with entry and exit, added 1.6 percentage points a year to productivity growth in 2004–2013 and 0.8 in 2013–2022, with the weakest windows in 2013–2016 and 2019–2022. By component, about half of the fall comes from entry and exit, mainly because the exit of weak firms contributed less, and about half from employment shifting less towards more productive firms. By ownership, about two-thirds of the fall comes from foreign-owned firms, which stopped gaining ground on less productive domestic firms and among which resources moved less. Among domestic firms alone, the contribution fell by about a quarter.

Exercise 3: the reserve is sizeable. The efficiency gap is about six percentage points of value added higher in 2016–2019 than in 2004–2010, and about ten points higher when 2020–2023 is included. The rise comes from wider gaps within sectors, almost entirely outside manufacturing, above all in construction and utilities, and it is at least as large among domestic firms as overall. Returns on capital became more unequal across firms before 2013 and stayed so. Returning to the 2004–2010 allocation would raise the level of output by about 1–2% of GDP, a one-off gain equivalent to 0.2–0.4 percentage points of extra growth a year over five years.

Table 3 summarises what each exercise measures, what it finds, and what it does not tell us.

Exercise 2 measures a flow: how much productivity growth comes from workers moving between firms and from firms entering and exiting. Exercises 1 and 3 measure a state: how far apart firms are, and how far allocation is from efficient, in a given year. The two can move differently, and they did.

Resources moved less, and misallocation became more costly

The flow weakened after 2013, most among foreign-owned firms, which also stopped gaining ground on less productive domestic firms; among domestic firms it slowed by about a quarter (Section 3.5).

The state did not improve, and on the model's measure it deteriorated. The efficiency gap rose, among domestic firms at least as much as in the economy as a whole, and most in construction and utilities. Returns on capital also became more unequal across firms, a rise that took place before 2013. Exercise 1 shows that the reallocation that did happen was not enough to bring Hungary's gaps down to Poland's level. Since we do not observe the reallocation flow in Poland, we cannot say whether resources moved faster there or whether less new dispersion was created.

Taken together, the exercises suggest that after 2013 workers moved less towards more productive firms, returns on capital remained more unequal across firms than before 2010, and the efficiency gap grew.

5 Policy discussion

Hungary has productive firms; they do not grow enough, and weak firms do not shrink enough

We started from the question of whether the Hungarian economy has resources that better policy could mobilise without new spending. Our results suggest that it does. The wide gaps of Exercise 1 mean that Hungary has highly productive firms in every industry. The problem is that they do not grow enough at the expense of less productive ones, and that less productive firms do not shrink or exit enough.

In an economy with gaps as wide as Hungary's, workers and capital would produce more in the firms that use them best. The benchmark above gives the order of magnitude at stake if allocation merely returned to its 2004–2010 level. Reaching it takes time, because moving resources involves adjustment costs, and it depends more on the overall policy environment than on any single measure.

Rules that treat firms unequally lower aggregate productivity

Our evidence does not identify which policy slowed reallocation, but the literature indicates which kinds of policy can. Restuccia and Rogerson (2008) show that taxes and rules that treat firms differently according to their identity rather than general criteria lower aggregate productivity, even when no firm becomes less efficient. Andrews and Cingano (2014) link allocative efficiency to product market regulation and barriers to entry. Andrews et al. (2017) show that weak firms surviving on soft financing crowd out productive ones. Within the model of Exercise 3, the rise in the efficiency gap does not come from labour: returns on labour did not become more unequal across firms. This does not rule out labour-market frictions behind the slower flow of workers in Exercise 2.

In the language of Exercise 3, anything that makes the effective tax rate differ across firms in the same industry adds to the gap: taxes and exemptions tailored

to particular firms or groups, discretionary subsidies and procurement, and corruption, which works like a tax that some firms pay and others do not. Using firm-level data from Uganda, Fisman and Svensson (2007) find that a one percentage point higher bribery rate is associated with about three percentage points lower firm growth, roughly three times the effect of taxation.

Retail: earlier work found reallocation weakened after 2010; it did not weaken further after 2013

Retail trade is a Hungarian example that has been studied in detail. Muraközy et al. (2018, Chapter 8) find that before the 2008 crisis, productivity growth in retail came mainly from reallocation, chiefly through entry and the expansion of large chains, many of them foreign-owned. This weakened after 2010, and the contribution of reallocation fell. They suggest that this may be related to the size-dependent measures introduced after 2010, such as the crisis taxes on retail. Our data show no further decline after 2013: the contribution of resources moving in retail stayed at 1.3–1.5 points a year, and its efficiency gap hardly changed (Section 3.6).

The Hungarian pattern is consistent with these mechanisms. Gaps between firms are wider than in Poland on every measure (Exercise 1 and Appendix D), returns on capital became more unequal across firms before 2013 and stayed so, and the efficiency gap grew most in construction and utilities but not in manufacturing, which is exposed to international competition (Exercise 3 and Section 3.6). Together, this points to an environment in which firms' growth depends on factors other than their productivity, although our evidence cannot rule out other explanations.

Better allocation does not mean the state choosing which firms grow

Our results could be read as a case for the state to pick productive firms and direct workers, capital or subsidies to them. They should not be. The benchmark of Exercise 3 is not a lower effective tax rate for productive firms but the same effective rate for every firm. Under equal rules, markets decide how large each firm becomes: firms that meet demand at low cost grow, and others shrink or exit. When the state lowers the effective rate for firms it selects, even firms that are productive today, it moves the economy away from this benchmark. The state cannot observe how large each firm should be: that information is dispersed across firms, customers and lenders, and it changes over time. Selected firms may then grow because of the favour rather than because of demand, and keep the favour after their advantage has gone. Other firms learn that seeking favours can pay more than improving. In Hungary, discretionary subsidies and individual agreements with selected companies on one side, and special taxes that fall on particular firms on the other, are the main forms of such differentiation (Section 3.5).

Subsidies should be assessed for their effect on allocation

This does not make better allocation free, and it does not rule out all support. Removing sector-specific levies costs revenue at a time when the deficit is the binding constraint, and changing the rules has political and adjustment costs, borne partly by workers in firms that shrink. Some support is also well justified. Where the benefits of an activity spill over to other firms or to society, as with research and development, training or cleaner production, or where firms with good projects find it hard to borrow, because they are young, have few fixed assets to pledge or run risky businesses, markets alone provide too little of it, and support can raise output. What our results add is a cost that is rarely

counted. Every subsidy, and every tax that singles out some firms, changes which firms grow: it shifts effective tax rates between competitors, and subsidies in particular can keep resources in firms that would otherwise shrink and make growth depend on access to the state. These effects are usually unintended, and they come on top of the direct cost to the budget. Support for a good reason is best given through criteria that any firm can meet, and the case for each measure should show that its benefits exceed both its budgetary and its allocative costs.

Equal, stable and transparent rules; temporary sector measures; open entry and exit

Four principles follow, none of which requires a new spending programme. First, rules should apply equally to all firms and be stable, so that firms grow because they are productive, not because they are favoured. This applies to foreign-owned firms too: some sector-specific taxes fell mainly on them and, according to Varju and Papp (2022), allowed discrimination against them, while nine in ten strategic cooperation agreements with the government were signed with foreign-owned companies (Section 3.5). Second, measures that change margins in particular industries, such as price caps and sector-specific taxes, should be temporary by design and reviewed for their effect on allocation. Third, entry and exit should be open. In our data most of the contribution of entry and exit comes from weak firms exiting, and this contribution fell from 1.3 percentage points a year before 2013 to 0.6 after. The literature on weak firms kept alive by soft financing (Andrews et al., 2017) describes one way this can happen; our data do not show whether it did. Fourth, support should be transparent: publishing the terms of individual agreements and subsidy decisions would allow their effect on allocation to be assessed.

Growth from better allocation does not require more investment

A final point concerns the composition of growth. Much of economic policy aims to increase the capital stock, including by supporting firms' investment. Reallocation works through a different channel: it raises total factor productivity, the output obtained from given inputs. The gain from better allocation therefore does not depend on support for capital accumulation, and it can be achieved even if that support is reduced. If capital and labour flow to the firms that use them best, the economy can grow faster during the transition with the same, or even a smaller, investment budget.

Recommendations

1. More equal treatment of firms means more growth from productivity

- If economic policy intervenes less in which firms grow, more of growth would come from total factor productivity, as workers and capital move to the firms that use them best.
- The stake is sizeable. Returning to the allocation of 2004–2010 would raise output by about 1–2% of GDP. This is a one-off rise in the level of output that then persists, equivalent to 0.2–0.4 percentage points of extra growth a year over five years.
- This gain needs no new spending programme and comes on top of what investment delivers. It can be achieved even if support for investment is reduced. The changes needed are not free: removing special levies costs revenue, and changing rules has political and adjustment costs.
- Better allocation does not mean the state choosing which firms grow. The benchmark is equal treatment, not a lower tax for firms the state judges to be productive.
- The gain comes gradually, because moving resources takes time, so it should be monitored. The gaps between firms, the contribution of resources moving and the efficiency gap can be updated every year from firm balance sheets; the

central bank's productivity report does not currently track them.

2. Policies that can help

Our evidence does not identify which policies slowed reallocation; the list draws on the research reviewed in this section and in Appendix A.

Stop or scale back

- Taxes, exemptions and special levies that apply to some firms or sectors but not to their competitors
- Discretionary subsidies and individual agreements that give selected firms preferential treatment
- Selecting firms to grow and channelling resources to them, however productive they look today
- Discretionary procedures in public procurement where open tendering is possible
- Price caps and margin regulations that remain in place after the emergency that justified them
- Soft financing, including state-backed lending, that keeps non-viable firms alive

Do more of

- Rules that apply equally to all firms, domestic and foreign-owned, and that remain stable
- A time limit and a review of every sector-specific measure, judged by its effect on allocation
- Open entry: fewer regulatory barriers, especially in services and network industries
- Faster exit: insolvency and restructuring rules that free up the workers and capital tied up in non-viable firms
- Where support is given, criteria that any firm can meet, for example for research and development, or credit access for firms that find it hard to borrow, such as young firms, firms with few fixed assets to pledge or firms with risky projects
- For every subsidy and special tax, an assessment that weighs its aims, such as spillovers from research, against its effect on which firms grow
- Publication of the terms of individual agreements and subsidy decisions, so that their effect on allocation can be assessed

6 Caveats

This section sets out the limits of our evidence.

The evidence is descriptive: it cannot show which policy caused the change

We document how allocation changed, but we cannot tell which policy, if any, caused the change. Each exercise also leaves questions open, as Table 3 shows. Gaps in revenue productivity can reflect prices, markups and measurement error as well as misallocation, and the model behind Exercise 3 provides a benchmark, not an estimate of what a reform would deliver.

The gap to Poland and the fall in reallocation are robust; trends and magnitudes less so

Some results are robust: the gap between Hungary and both Poland and the old member states in the late 2010s; the fall in the contribution of resources moving between 2004–2013 and 2013–2022, which holds in every specification we tried; and the sectoral pattern of the efficiency gap. The efficiency gap is large in every specification, but its level remains a model benchmark.

Other results are less robust. The contrast in trends between Hungary and Poland depends on the measure, and so does any split between labour and capital (Appendix D). The contribution of resources moving is volatile from window to window, and the size of its decline differs across methods (it is larger with the Melitz–Polanec decomposition). The size of the reserve depends on the model's assumptions and on the period compared, and the pre-pandemic comparison depends on trimming. The ownership data change source in 2015

and again after 2020, so the split by ownership is indicative. Finally, a similar rise in the efficiency gap has been reported for Poland, so the rise over time cannot be read as specific to Hungary.

2020–2023 were crisis years; the 2024 data will be added in the next update

Several results rely on 2020–2023: the weak reallocation window of 2019–2022 and much of the rise in the efficiency gap. These were the years of the pandemic, the energy shock and the price caps, and the 2023 data are provisional. Data for 2024 will be added in the next update. They will help show whether the weakening persists once the crisis years have passed, or whether a recovery follows, as it partly did after 2013–2016.

Our results are therefore consistent with workers moving less towards more productive firms, especially among foreign-owned firms, with returns on capital becoming more unequal, and with the cost of misallocation rising, at least as much among domestic firms as overall. But they do not prove it.

Technical appendix

These appendices review the related literature, list data sources and sample rules, give the formulas, explain the choices made and the alternatives considered, and report robustness checks and the full ranges behind the magnitudes in the main text. The code is organised in three automated pipelines, one per exercise, each with a decision log (Appendix H).[7]

A Related literature

The note draws on six strands of research: the measurement of misallocation, the critiques of that measurement, the decline of reallocation in advanced economies, evidence from Europe and the region, work on Hungary, and studies that link specific policies to allocative efficiency.

Misallocation and the efficiency gap. Restuccia and Rogerson (2008) and Hsieh and Klenow (2009) started the modern literature. Restuccia and Rogerson show in a model that policies which tax some firms and subsidise others lower aggregate productivity even when no firm becomes less efficient. Hsieh and Klenow measure the dispersion of marginal revenue products across manufacturing plants in China and India and compute how much output would rise if this dispersion were removed; this is the calculation of Exercise 3. Restuccia and Rogerson (2017) and Hopenhayn (2014) review the causes and the theory. Bartelsman et al. (2013) show that the within-industry covariance between firm size and productivity, a measure of allocative efficiency, differs widely across countries. The decomposition of Exercise 2 goes back to Baily et al. (1992) and Griliches and Regev (1995); we use the version of Foster et al. (2001) and check it with Melitz and Polanec (2015). Foster et al. (2008) show that revenue and physical productivity can differ, which matters because we observe only revenue.

What dispersion does and does not show. A wide spread of revenue productivity need not mean misallocation. Haltiwanger et al. (2018) show that it can also reflect differences in demand and technology that the standard model leaves out. In nine datasets covering 40 countries, Asker et al. (2014) find that an investment model with adjustment costs and volatile productivity accounts for 80–90% of the variation in the dispersion of returns to capital across industries and countries. David and Venkateswaran (2019) split the dispersion of returns to capital into its sources: in Chinese manufacturing, adjustment costs and uncertainty explain only a modest share, and most of it comes from firm-specific factors correlated with productivity, the kind of wedge that policy can create. Bils et al. (2021) correct for measurement error in revenue and inputs; the correction lowers the estimated gain from reallocation by 20% in India and by 60% in US manufacturing. Using an EU-wide firm survey, Gorodnichenko et al. (2026) find that part of the measured dispersion reflects firm characteristics and may compensate for differences between firms rather than reflect distortions. For this note, these studies imply that the level of the Exercise 3 gain should be read as a benchmark rather than an estimate, and that its change over time is more informative than its level.

Declining reallocation and divergence. In the United States, Decker et al. (2017) link the productivity slowdown to weaker allocative efficiency, and Decker et al. (2020) show that firms have become less responsive to their own productivity: productive firms grow less than they used to. This is the closest parallel to the fall in Exercise 2 and to Koren and Rátfai (2026) for Hungary. In OECD data, Andrews et al. (2016) find that the productivity gap between frontier firms and the rest widened, more so in sectors where pro-competitive product market reforms were less extensive, and Berlingieri et al. (2017) find that productivity dispersion rose within sectors in 16 countries, driven

[7] This is a draft. Work in progress includes: (i) regressions of firms' employment and capital growth on their initial productivity, by period, to test directly whether resources respond less to productivity than before; (ii) capital-only and labour-only versions of the Hsieh–Klenow calculation, to test within the model whether the rise in the efficiency gap comes from capital; (iii) a sample bridge showing the number of firms, employment, value added and capital remaining after each filter; (iv) dispersion with fixed factor weights on the Hungarian data, and a decomposition of the cost-share measure into the dispersion of labour productivity, of capital intensity and their covariance; (v) an update of the regional comparison using the 10th vintage of CompNet; and (vi) generating all numbers in the text directly from the output files.

mainly by the bottom of the distribution (published as Berlingieri et al., 2024). Adalet McGowan et al. (2018) and Andrews et al. (2017) show that the share of capital sunk in zombie firms, old firms that persistently struggle to cover their interest payments, rose from the mid-2000s, and that this is associated with lower investment and employment growth among other firms in the same industry and weaker reallocation of capital. Andrews and Petroulakis (2019) link zombie firms in Europe to weak banks.

Europe and the region. CompNet, the source of our comparisons, is described by Lopez-Garcia et al. (2015) and CompNet (2021). Using firm data for five euro-area countries in 2002–2012, Gamberoni et al. (2016) find that the allocation of capital worsened everywhere except Germany while the allocation of labour changed little, and that the deterioration was larger in services than in industry. This is the pattern of our Exercise 3. Calligaris et al. (2018) estimate that Italy's aggregate productivity in 2013 would have been 18% higher had misallocation remained at its 1995 level; the rise occurred within sectors. Dias et al. (2016) find that misallocation within industries in Portugal almost doubled between 1996 and 2011, concentrated in services and on the capital side. Gopinath et al. (2017) link the rise in capital misallocation in Spain to the fall in interest rates after the introduction of the euro. For the region, Giordano and Lopez-Garcia (2018) use CompNet data for nine Central and Eastern European countries, Hungary and Poland among them, for 2003–2012, and find that more corruption is associated with more misallocation of both labour and capital, especially where regulation is of lower quality. Grassi (2026) shows that the 2004 joiners narrowed the productivity gaps between their firms towards the level of the old member states, and Gradzewicz (2026) applies the Hsieh–Klenow calculation to Poland. At the level of the EU, the Draghi report attributes the productivity gap with the United States largely to the technology sector (Draghi, 2024); this note looks at a different margin, allocation within industries.

Hungary. Before ours, the most detailed decomposition of Hungarian productivity growth is that of Muraközy et al. (2018), who find strong reallocation before 2008 and much weaker reallocation after 2013. Harasztosi (2011) decomposes Hungarian growth in 1994–2008 into the contributions of capital, labour, productivity and reallocation. Koren and Rátfai (2026) find that the employment growth advantage of the most productive Hungarian firms over the median firm has shrunk steadily since the 1990s, and that the growth advantage of foreign-owned firms over non-exporting domestic firms has disappeared. Recent OECD surveys reach a similar diagnosis. OECD (2024), citing Muraközy et al. (2018), reports that the within-firm, between-firm and entry and exit contributions to productivity growth have all declined since the mid-2000s. OECD (2026) points to low business dynamism, product market regulation that holds back entry, insolvency rules that slow the reallocation of capital towards the most productive firms, and concentration and markups that have risen faster than elsewhere in Europe. The central bank's productivity report documents a large productivity gap between foreign-owned and domestic firms, about threefold in 2010–2016 and two-and-a-half-fold in 2021, but does not analyse allocation (Magyar Nemzeti Bank, 2024).

Several studies show how public resources were allocated across Hungarian firms. Exploiting a reform that allowed a more discretionary procedure below a contract-value threshold, Szucs (2024) finds that discretion in public procurement raised prices, led to the selection of less productive contractors and favoured firms connected to the governing party. Muraközy and Telegdy (2016) find that EU-funded grants in 2004–2012 favoured municipalities aligned with the governing coalition mainly for projects visible to voters, such as construction. Muraközy and Telegdy (2023) and Banai et al. (2020) find that EU grants to small and medium-sized firms raised employment and sales. In the first study labour productivity also rose, together with capital per worker, but the results for TFP are conflicting, which the authors read as more investment with little technology upgrading; in the second, labour productivity did not change. Varju and Papp (2022) analyse the sector-specific special taxes from a legal perspective and argue that their design allowed discrimination, in particular against foreign-owned firms.

Policies and allocative efficiency. Andrews and Cingano (2014) link allocative efficiency to product market regulation, employment protection and barriers to entry, and Arnold et al. (2008) find that restrictive regulation of services goes with less efficient allocation within industries. Garicano et al. (2016) quantify one size-dependent rule: French labour regulations that apply from 50 employees act like a 2.3% tax on labour and cost 1.3–3.4% of GDP,

depending on how flexible wages are. Midrigan and Xu (2014) find that financial frictions are costly mainly because they reduce entry and technology adoption, not because they cause misallocation among incumbents. Fisman and Svensson (2007) find that bribery hurts firm growth more than taxation does. We are not aware of any study that links price caps to the allocation of resources across firms.

What this note adds. For Hungary, the note extends the evidence on productivity dispersion and on the sources of productivity growth to 2023, compares Hungary with Poland and the region on comparable measures, and separates the contributions of foreign-owned and domestic firms.

B Data

Source. Mérleg LTS, version 5.9.0 (November 2025), compiled by KRTK from corporate tax returns up to 2014 and Opten company records from 2015 onwards, with contributions from CEU MicroData. The panel contains one record per firm and year for 1980–2023, 11.0 million firm-years in total; we use 2004–2023. The firm identifier (`frame_id`) follows firms through mergers and transformations; where it is missing, we use the tax number. Industry codes are harmonised to two-digit NACE Rev. 2 and are year-specific, so a firm's industry can change over time.

Variables. Value added: sales plus capitalised own work minus material costs (`gdp`); for the dispersion measure in Exercise 1, we follow CompNet's definition, sales minus material costs. Labour: number of employees (`emp`). Labour cost: payments to personnel including employer contributions (`persexp`). Capital: tangible plus intangible fixed assets, book value. Capital cost: depreciation (`kecs`); in the version with an imputed financing cost, we add the 10-year government bond yield (Eurostat series `irt_lt_mcby_a`) multiplied by the sum of equity (`satok`, if positive) and long-term liabilities (`hlk`). Exercises 2 and 3 use this version. All values are in thousand HUF, nominal unless otherwise stated; Exercise 2 deflates them as described under Prices.

Sample for Exercise 1. Business entities (company-form codes 1xx, which exclude sole proprietors, as in CompNet), with at least 20 employees, in CompNet's industries (NACE 10–33 excluding 19, 41–43, 45–47, 49–53, 55–56, 58–63, 68, 69–75, 77–82), with positive sales, value added, capital and material costs, 2004–2023. CompNet's outlier routine is applied before the size threshold.

Sample for Exercises 2 and 3. Market economy (NACE 05–82 and 95, excluding 53 and 64–66); firms that were ever non-profit or budgetary are dropped. There is no company-form filter. An active firm-year has at least one employee and positive value added and capital (for Exercise 3, also a positive wage bill). For each window of Exercise 2, we keep firms with at least 5 employees in at least one end-year of the window. For Exercise 3, we keep firm-years with at least 5 employees. In Exercise 2, productivity is winsorised at the 1st and 99th percentiles by industry and year, and, following Muraközy et al. (2018), firms whose log productivity changes by more than 1.5 between consecutive years with 5 or more employees are dropped. Exercise 3 instead trims 1% at each tail (Appendix G). Each three-year window contains 34,000–58,000 continuing firms and 8,000–16,000 entrants and exiters.

Coverage. In every year from 2015 to 2022, the nominal value added of the Exercise 2 sample is 62–64% of national-accounts gross value added in the same industries, and it grows at almost the same rate (2022: 18.3% against 18.4%). In 2023, however, it grows by 5.9% against 15.9% in the national accounts. The number of active firms does not fall (180,755 in 2023 against 179,520 in 2022), but this does not rule out changes in composition. The source of the gap is unknown, so we treat 2023 as provisional.

Prices. Exercise 2 uses value-added price indices in the main specification and producer price indices as an alternative. We take Eurostat value-added price indices by A64 industry and the price index for gross fixed capital

formation (nama_10_a64, nama_10_gdp); for air transport, we use the index for transport as a whole. Producer price indices are included in the panel. A deflator common to all firms in an industry and year cancels out of the coefficient of variation in Exercise 1 and out of the within-industry efficiency ratio in Exercise 3, but firm-specific prices and markups do not.

CompNet. We use the 8th vintage: firms with 20 or more employees, unweighted, at the two-digit industry level, aggregated to country-year by Basile Grassi. Data for Hungary start in 2004 and for Poland in 2002, and run to 2019 (2018 for most old member states). The CEE average covers the six 2004 joiners with available data, including Hungary. The old member states are the nine EU-15 countries with available data, shown for years in which at least five report. We examined the 9th vintage but did not use it, because it no longer reports the fixed-weight residual, one of our three measures.

GDP and deficit. Nominal GDP 2023: HUF 75,218 billion (Eurostat nama_10_gdp, B1GQ, current prices). General government balance: −7.0% of GDP in 2023, −5.1% in 2024, −4.7% in 2025 (Eurostat gov_10dd_edpt1, April 2026 release). Forecasts for 2026: −7.5% (Ministry of Finance, July 2026 budget review, the basis of the revised 2026 budget submitted in August; Pénzügyminisztérium, 2026) and −6.9% (Magyar Nemzeti Bank, 2026, September 2026 Inflation Report).

C Exercise 1: method and validation

Firm productivity. For firm i in industry s and year t,

$$SR_{ist} = \frac{VA_{ist}}{L_{ist}^{\alpha_{st}} K_{ist}^{1-\alpha_{st}}}, \tag{1}$$

where α_{st} is the median, across firms in the industry and year, of labour cost divided by the sum of labour cost and capital cost. This imposes constant returns to scale. In the baseline, capital cost equals depreciation. In the second version, it is depreciation + r_t (equity + long-term debt), where r_t is the nominal 10-year bond yield; this is an accounting proxy for CompNet's capital cost, which is depreciation plus interest paid plus an imputed interest on equity (CompNet, 2021).

Outliers. CompNet's routine (CompNet, 2021): a firm-year is dropped if any of six ratios (sales to labour, to labour cost, to capital and to materials; capital to labour; materials to labour) falls in the bottom 3% or top 2% of its industry-year distribution, computed on all firms before the 20-employee cut. This reproduces CompNet's firm counts within 2.5% in 2004–2015.

Dispersion and aggregation. For each industry-year with at least 10 firms, $CV_{st} = \mathrm{sd}(SR)/\mathrm{mean}(SR)$. The yearly value is

$$CV_t = \Big(\sum_s N_{st}\, CV_{st}^2 \Big/ \sum_s N_{st} \Big)^{1/2},$$

with N_{st} the number of firms (Grassi's aggregation). Regional averages apply the same formula across countries.

Period averages. On our Hungarian series, the spread is 0.78 in 2005–2008, 0.73 in 2016–2019, 0.72 in 2020–2023 and 0.70 in 2023 alone. The corresponding values for 2005–2008 and 2016–2019 are 0.80 and 0.62 for Poland, 0.76 and 0.64 for the CEE average, and 0.58 and 0.57 for the old member states. In 2019, the Hungarian value was 0.15 above the Polish one (CompNet's own Hungarian series: 0.69 against 0.58). The version with imputed financing cost is 0.04–0.06 above the baseline in 2009–2013 and 2022–2023.

The implied composite coefficient. CompNet reports mean log labour productivity $E[\ln(VA/L)]$ and mean log Solow residuals with fixed (2/3) and cost-share weights. Since $\ln SR = \ln(VA/L) - (1-\alpha)\ln(K/L)$,

$$E[\ln(K/L)] = 3\left(E[\ln(VA/L)] - E[\ln SR^{2/3}]\right), \qquad \bar{\alpha} = 1 - \frac{E[\ln(VA/L)] - E[\ln SR^{cs}]}{E[\ln(K/L)]}. \tag{2}$$

With industry-specific weights, $\bar{\alpha}$ is a composite, $E[\alpha_s \ln(K/L)]/E[\ln(K/L)]$, rather than the average labour share, and it depends on the units of K/L. We therefore use it only as a diagnostic. For Hungary it rises from 0.75 (2004) to 0.84 (2019), for Poland from 0.76 to 0.86. On our data the composite is 0.004–0.007 below the mean weight.

Table 4: Our Hungarian series against CompNet's, 2004–2019

Labour weight	CompNet series	Mean gap	RMSE	Correlation	Labour-weight gap
Fixed weights, labour 2/3	PV08	−0.009	0.026	0.83	–
LC / VA	PV09	0.084	0.100	0.16	−0.10 [−0.18, −0.05]
LC / (LC + depreciation)	PV09	0.002	0.021	0.86	0.07 [0.03, 0.09]
LC / (LC + depr. + imputed financing cost)	PV09	0.024	0.037	0.75	−0.02 [−0.04, −0.01]

Notes: Gap = our value minus CompNet's. PV08: CompNet's Solow residual with fixed weights; PV09: with cost shares. Labour-weight gap: our implied composite coefficient minus CompNet's, mean and [range]. Number of firms within 2.5% of CompNet's in every year 2004–2015.

Table 4 shows the check. With fixed weights, our series closely matches CompNet's (mean gap −0.009), which validates the sample and the data. With cost shares, weights based on labour cost over value added do not reproduce CompNet's series (correlation 0.16); the two versions we report are within 0.002 (baseline) and 0.024 (imputed financing cost) of CompNet's on average. The financing-cost version comes closer to CompNet's implied labour weights, but the baseline reproduces CompNet's spread more closely, so we use the baseline as our main measure. After 2015, our firm count grows while CompNet's stays flat, and our series lies slightly above CompNet's (2019: 0.73 against 0.69). We have not established why; the most likely cause is our switch from tax returns to Opten company records in 2015.

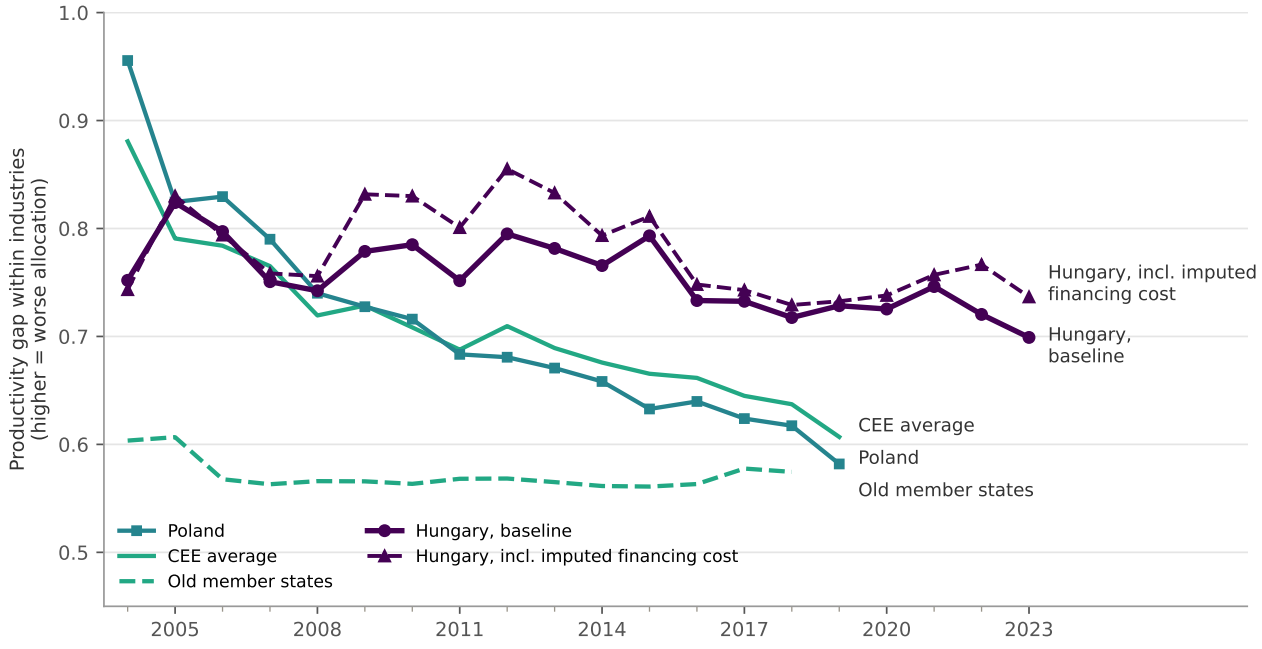


Figure 6: Exercise 1 with an imputed financing cost in the cost of capital, 2004–2023
Notes: As Figure 1, with a second Hungarian line in which the cost of capital also includes the 10-year government bond yield multiplied by the sum of equity (if positive) and long-term debt. The second line is 0.04–0.06 higher in 2009–2013 and 2022–2023, when interest rates were high, and close to the baseline in other years. The CompNet series are unchanged.

D Exercise 1 on three productivity measures

Productivity can be measured in more than one way, and the weights given to labour and capital are a choice. This appendix checks whether the comparison of Exercise 1 is robust to that choice. We compute the same spread with three productivity measures that CompNet publishes: labour productivity (value added per employee), TFP with fixed weights (two-thirds labour, one-third capital), and TFP with cost-share weights, the measure used in Exercise 1. Each CompNet indicator uses the firms that have the variables it needs. We use CompNet's own series for Hungary, Poland, the CEE average and the old member states, and compare the averages of 2005–2008 and 2016–2019 (CompNet, 2021); Figure 7 shows the yearly series.

Table 5: Three CompNet measures of productivity gaps, period averages

	A. TFP, cost-share weights			B. TFP, fixed weights			C. Labour productivity		
	2005–08	2016–19	Change	2005–08	2016–19	Change	2005–08	2016–19	Change
Hungary	0.78	0.71	−0.08	0.83	0.83	0.00	0.85	0.73	−0.12
Poland	0.80	0.62	−0.18	0.73	0.73	0.00	0.74	0.62	−0.12
CEE average (2004 EU joiners)	0.76	0.64	−0.13	0.76	0.75	−0.01	0.76	0.65	−0.11
Old member states (EU-15)	0.58	0.57	0.00	0.77	0.70	−0.06	0.59	0.60	0.01

Notes: Standard deviation divided by the mean within two-digit industries, aggregated across industries as a root mean square weighted by the number of firms, and across countries in the same way; CompNet 8th vintage, firms with 20 or more employees, all from CompNet's own series (Hungary is CompNet's Hungarian line, slightly different from ours). Old member states (EU-15): the nine EU-15 countries with data; 2016–2018. Fixed weights: labour 2/3, capital 1/3. Changes are computed from unrounded values.

Levels. Table 5 gives the levels. In 2016–2019, Hungary's gaps were wider than Poland's on all three measures, by 0.09–0.11, and wider than the CEE average by 0.07–0.08; since the CEE average includes Hungary, the second comparison is conservative. Poland's gaps were close to those of the old member states on all three measures, although the old member states' own level depends on the measure (0.57–0.60 on cost-share TFP and labour productivity, 0.70 on fixed-weight TFP). The difference in levels between Hungary and the others does not depend on how productivity is measured.

Trends. How the difference arose does depend on the measure. On the cost-share measure Hungary and Poland started at about the same level in 2005–2008, and Poland's gaps narrowed much more. On labour productivity Hungary's gaps were already about 0.1 wider in 2005–2008, and both countries narrowed by the same 0.12. With fixed weights neither country changed. So we do not read the comparison as Hungary falling behind Poland. Nor do we take a view on whether the difference lies with labour or with capital: the measures weight the two differently, and the results move with the weights.[8]

Hungary after 2019. CompNet's data end in 2019, so Poland cannot be followed further. For Hungary we can extend all three measures with our own data (Table 6); our labour-productivity series closely follows CompNet's in 2004–2019 (mean gap 0.002, correlation 0.91), though, as in Appendix C, it is slightly above it after 2015. Between 2016–2019 and 2020–2023, the gaps hardly changed: from 0.73 to 0.72 on the cost-share measure, from 0.81 to 0.80 with fixed weights, and from 0.76 to 0.75 on labour productivity. The narrowing on labour productivity before 2019 did not continue. This has two implications. First, the gap to Poland and the old member states shown by Exercise 1 for 2019 did not close afterwards on the Hungarian side; we cannot tell whether Poland's gaps moved. Second, the shocks of 2020–2023 (the pandemic, the energy shock and the price caps) could have pushed resources towards better firms or kept them where they were. On these measures the gaps neither narrowed nor widened, so by this yardstick allocation did not improve.

[8] The cost-share measure weights labour and capital by their shares in costs. In CompNet's data, the implied labour weight (the composite coefficient of Appendix C) rose from about 0.75 to 0.85 in both countries between 2004 and 2019, partly because falling interest rates lowered the cost of capital. Changes in this measure therefore mix changes in the data with changes in the weights, and CompNet's aggregates cannot separate the two. Our own baseline uses depreciation only as the cost of capital, so this interest-rate channel does not affect it.

Table 6: Productivity gaps in Hungary on three measures, own data, 2005–2023

	2005–08	2016–19	2020–23	Change to 2016–19	Change to 2020–23
TFP, cost-share weights (Exercise 1)	0.78	0.73	0.72	−0.05	−0.01
TFP, fixed weights (labour 2/3)	0.81	0.81	0.80	0.00	0.00
Labour productivity	0.84	0.76	0.75	−0.08	−0.01

Notes: Standard deviation divided by the mean within two-digit industries, aggregated as in Figure 1. Firms with 20 or more employees in CompNet's industries, own calculation from the Mérleg LTS panel. Cost-share weights: labour cost over labour cost plus depreciation, the Hungarian line of Figure 1. CompNet data, and hence Poland, end in 2019. With 2021–2023 instead of 2020–2023 the values are the same to two decimals. Changes are computed from unrounded values.

Summary. Taken together, the three measures agree that firms in the same industry are further apart in Hungary than in Poland and the region, and that in Hungary these gaps did not narrow after 2019. They do not agree on when the difference arose, or on whether it lies with labour or with capital. Decomposing the cost-share measure, holding the weights fixed, into the dispersion of labour productivity, the dispersion of capital intensity and their covariance would separate changes in the data from changes in the weights. The decomposition is exact for the variance of logs and only approximate for the coefficient of variation we use. It can be done on the Hungarian firm data, but not on CompNet's aggregates (work in progress, item (iv)).

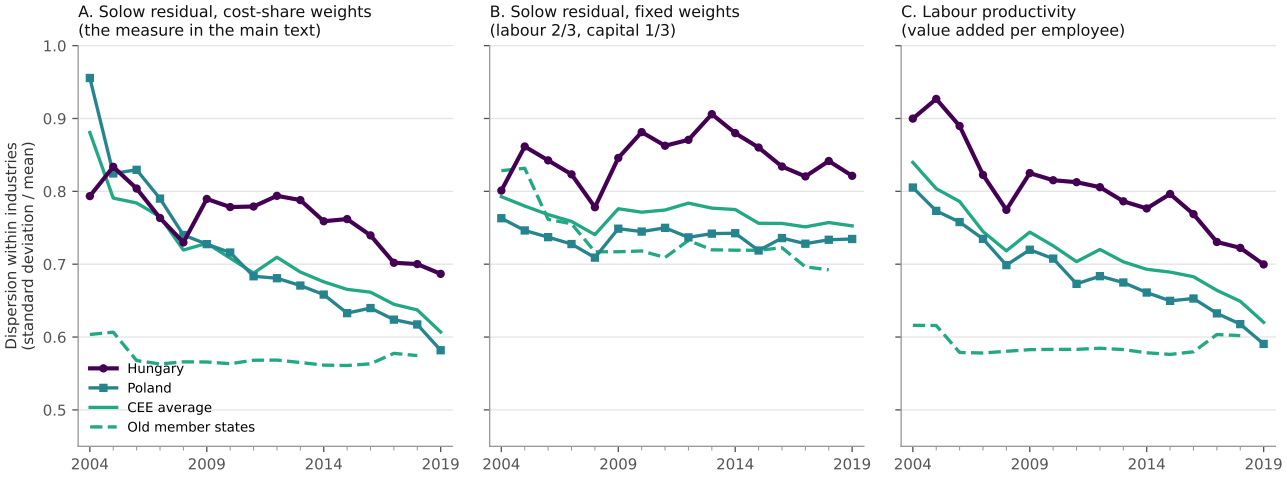


Figure 7: Three CompNet measures of productivity gaps, 2004–2019
Notes: As Table 5, yearly values. Hungary is CompNet's own line.

E Exercise 2: decomposition method and comparison with Muraközy et al. (2018)

Productivity. For firm i in industry s and year t, $tfp_{it} = \ln VA_{it} - \alpha_s \ln L_{it} - (1-\alpha_s) \ln K_{it}$, where α_s is the median, over 2004–2023 and across firms with 5 or more employees in industry s, of labour cost divided by the sum of labour cost, depreciation and the imputed financing cost (median across industries 0.78, range 0.17–0.97). Unlike in Exercise 1, the weights are fixed over time, so that productivity growth is not affected by changes in factor shares, and they include the financing cost to reflect the full cost of capital. Value added is deflated with the industry value-added price index and capital with the gross fixed capital formation deflator; the producer-price version deflates both with the industry producer price index.

Decomposition. For industry s between years a and b, with employment shares θ_{it} and $P_t = \sum_i \theta_{it}\, tfp_{it}$,

$$\begin{aligned} P_b - P_a = & \sum_{i\in C} \theta_{ia}\, \Delta tfp_i && \text{(within)} \\ & + \sum_{i\in C} (tfp_{ia} - P_a)\, \Delta\theta_i && \text{(between)} \\ & + \sum_{i\in C} \Delta tfp_i\, \Delta\theta_i && \text{(cross)} \\ & + \sum_{i\in N} \theta_{ib}\, (tfp_{ib} - P_a) && \text{(entry)} \\ & - \sum_{i\in X} \theta_{ia}\, (tfp_{ia} - P_a) && \text{(exit)} \end{aligned} \tag{3}$$

where C, N and X are the continuing, entering and exiting firms *of industry* s. Reallocation among continuing firms is the between plus the cross term. A firm active in both years but in different industries counts as an exiter from one industry and an entrant into the other: 765–5,578 firms per window, or 17–41% of entrants and 17–35% of exiters. The terms add up exactly, which the code verifies. Industry terms are aggregated using the average of each industry's employment share in a and b. The windows are 2004–07, 2007–10, 2010–13, 2013–16, 2016–19, 2019–22 and 2022–23. Values are divided by window length, and period values are averages weighted by window length.

The cross term. Since productivity contains $-\alpha_s \ln L$, a firm that adds workers before its output follows, or whose employment is mismeasured, shows productivity and employment share moving in opposite directions. A negative cross term is therefore consistent with resources not following productivity gains, but also with adjustment lags, labour hoarding in a crisis and measurement error. As a sensitivity check, we split the cross term equally between the within and between terms: within + cross/2 and between + cross/2.

Comparison with Muraközy et al. (2018). Their Figure 6.3 shows the same decomposition for 2004–2016. Using their sample and pooling all industries, as they do, and measuring capital in the same units, our 2004–2010 reallocation term is 1.50, against their 1.48. Our main version differs from theirs in four ways. (i) We decompose within industries. With industry-specific weights, TFP levels of different industries are on different scales: measuring capital in million instead of thousand forints shifts each firm's TFP by $(1 - \alpha_s) \ln 1000$. Within-industry results are unaffected, while the pooled 2004–2010 reallocation term falls from 1.50 to 0.97. Decomposing within industries makes the results independent of units. (ii) Entry and exit are defined by presence in an industry's sample, not by crossing the 5-employee threshold. (iii) We use cost-share weights rather than estimated production functions (Ackerberg et al., 2015). (iv) Our data run to 2023, and we report results with two price indices.

Why employment shares. With value-added TFP, value-added shares and the productivity measure contain the same measurement error in value added, which inflates the cross term (by 3.5 percentage points a year in 2004–2010 in our data). We therefore weight firms by employment. Employment shares are not free of this problem, but the bias works in the opposite direction and is scaled by the labour weight (see the cross term above).

F Exercise 2: robustness

Table 7 reports six specifications and the equal split of the cross term; Figure 8 shows the reallocation term alone. In every specification, the contribution of resources moving is lower in 2013–2022 than in 2004–2013, and lower in 2019–2022 than in 2016–2019. Outside Melitz–Polanec, the decline between the two periods ranges from 0.7 to 1.0 percentage points a year. Dropping firms that change industry raises the early contribution (1.80 in 2004–2013) and leaves the later one nearly unchanged (0.79). Splitting the cross term equally raises both levels but leaves the decline about the same size (1.85 to 1.14). With the Melitz–Polanec decomposition the contribution turns negative after 2013 (−0.10 a year, and −1.15 in 2019–2022), because the size–productivity covariance among surviving

Table 7: Robustness of the decomposition: resources moving (reallocation + entry and exit), annual percentage points

	04–13	13–22	13–16	16–19	19–22	13–23
Main: within industries, value-added deflators	1.58	0.84	0.68	1.18	0.65	0.67
Producer price index deflators	1.68	0.77	0.80	1.15	0.36	0.66
Labour weight: LC / (LC + depreciation)	1.54	0.80	0.57	1.21	0.63	0.61
Entry = crossing 5 employees	1.81	1.07	0.73	1.48	1.00	0.89
Firms changing industry dropped	1.80	0.79	0.72	1.06	0.60	0.64
Melitz–Polanec (2015)	2.02	−0.10	0.47	0.37	−1.15	−0.19
Main, cross term split equally (between + cross/2 + entry and exit)	1.85	1.14	0.90	1.38	1.13	1.01

Total productivity growth, same specifications

	04–13	13–22	13–16	16–19	19–22	13–23
Main: within industries, value-added deflators	2.14	3.67	2.53	5.13	3.36	2.43
Producer price index deflators	2.34	2.95	3.41	5.09	0.35	2.55
Labour weight: LC / (LC + depreciation)	2.22	3.65	2.65	5.38	2.92	2.41
Entry = crossing 5 employees	2.20	3.67	2.51	5.14	3.37	2.44
Firms changing industry dropped	2.29	3.72	2.64	5.15	3.36	2.47
Melitz–Polanec (2015)	2.14	3.67	2.53	5.13	3.36	2.43

Notes: 04–13 and 13–22 are averages of three windows each; 13–16, 16–19 and 19–22 are single windows; 13–23 adds the provisional one-year window 2022–23, weighted by window length. Labour weight: cost shares without the financing cost. Entry = crossing 5 employees: the Muraközy et al. (2018) sample, firm-years with 5 or more employees, so crossing the threshold counts as entry or exit. Firms changing industry dropped: such firms are removed from both ends of each window. Melitz–Polanec: reallocation is the change in the size–productivity covariance among surviving firms, and entry and exit are measured against survivors. The Melitz–Polanec and cross-term-split versions leave total productivity growth unchanged: the Melitz–Polanec row of the lower panel repeats the main row, and the cross-term split has no row there.

firms fell; that decomposition measures entrants against survivors rather than against the industry average, so its terms are not directly comparable with the main ones. Figure 9 shows all components with both price indices. Total growth depends on the price index after 2019: in 2019–2022, productivity grows by 0.4% a year with producer prices and by 3.4% with value-added prices.

The 2022–2023 window. With value-added deflators, this one-year window shows within-firm growth of −7.9 percentage points and total growth of −8.8; with producer price indices, −0.7 and −1.1. The national-accounts value-added deflator rose by 16–19% in 2023, while our firms' nominal value added grew by only 6%, so deflating by it implies a sharp fall in real value added. Under both deflators the contribution of resources moving is negative in this window (−0.9 points with value-added deflators, −0.4 with producer prices), which is what pulls the 2013–2023 column of Table 1 below 2013–2022. The between and exit terms are nearly the same under both deflators; the cross and entry terms differ. The problem therefore affects mainly the within-firm term and total growth. Because the window is provisional, we do not use it for the main finding on reallocation.

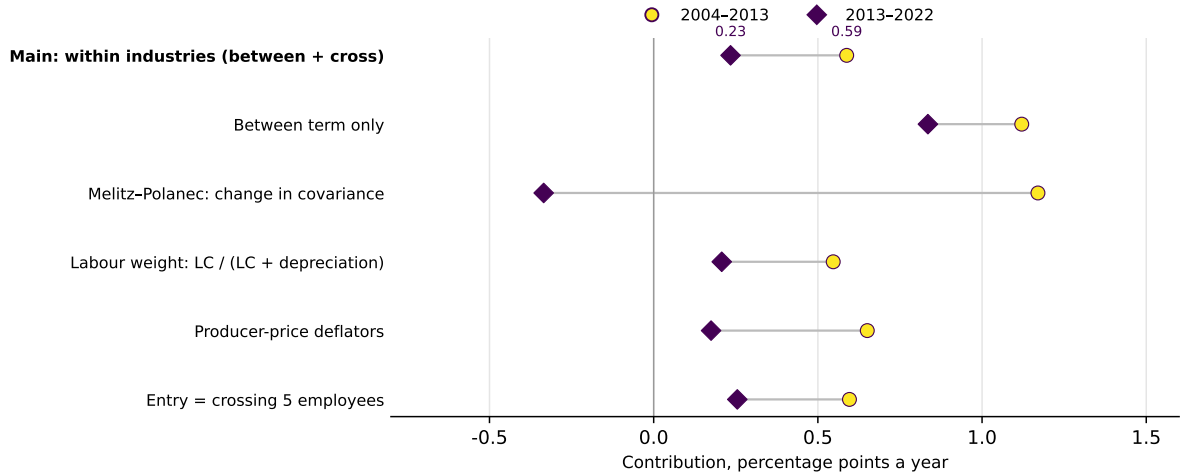


Figure 8: Reallocation among continuing firms by method, before and after 2013

Notes: Reallocation term only, excluding entry and exit (between + cross; for Melitz–Polanec, the change in the size–productivity covariance among surviving firms), by specification, in percentage points a year. Averages of the three-year windows in 2004–2013 and 2013–2022. Selected specifications from Table 7; the figure also shows the between term alone, which is not reported in Table 7. Entry = crossing 5 employees: firm-years with 5 or more employees, as in Muraközy et al. (2018).

G Exercise 3: Hsieh–Klenow calculation

Method. Following Hsieh and Klenow (2009, Section IV), for firm i in two-digit industry s and year t, with value added PY_i, capital K_i, wage bill wL_i and capital share α_s:

$$TFPR_i = \frac{PY_i}{K_i^{\alpha_s}(wL_i)^{1-\alpha_s}}, \qquad A_i \propto \frac{PY_i^{\sigma/(\sigma-1)}}{K_i^{\alpha_s}(wL_i)^{1-\alpha_s}},$$

$$\overline{TFPR_s} = \frac{\sum_i PY_i}{\left(\sum_i K_i\right)^{\alpha_s}\left(\sum_i wL_i\right)^{1-\alpha_s}}, \qquad A_s = \left(\sum_i A_i^{\sigma-1}\right)^{\frac{1}{\sigma-1}},$$

$$\frac{TFP_s}{TFP_s^{\text{eff}}} = \left[\sum_i \left(\frac{A_i}{A_s}\frac{\overline{TFPR_s}}{TFPR_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}},$$

$$\frac{Y^{\text{eff}}}{Y} = \prod_s \left(\frac{TFP_s^{\text{eff}}}{TFP_s}\right)^{\theta_s},$$

with θ_s the industry's share in value added. The efficiency gap is $100\,(Y^{\text{eff}}/Y - 1)$, computed year by year in nominal terms, with $\sigma = 3$ as in Hsieh and Klenow. In the model, a firm's $TFPR$ is high when it faces a tax-like distortion that keeps it small, and low when it is favoured and too large. Misspecified technology, measurement error and demand assumptions also create $TFPR$ dispersion (Haltiwanger et al., 2018), so the efficiency gap is a benchmark from a frictionless model, not a bound on the true gain in either direction.

Choices. We set $\alpha_s = 1 - \text{median}\left[wL/(wL + D + r_t E)\right]$ over firm-years with 5 or more employees in the industry, 2004–2023 (median across industries: 0.22), where D is depreciation and E is equity plus long-term debt. This is the same weight as in Appendix E: following Hsieh and Klenow, α_s here denotes the capital share, one minus the labour weight denoted α in Appendices C and E. Hsieh and Klenow use US industry shares, which are not available at our level of detail. We use two-digit industries, whereas they use four-digit ones; coarser industries usually raise the measured gain, so our levels are not comparable with theirs. Following them, each year we trim the top and bottom 1% of $\ln(TFPR_i/\overline{TFPR_s})$ and of $\ln(A_i M_s^{1/(\sigma-1)}/A_s)$, where M_s is the number of firms in the industry, and recompute the industry aggregates. Industry-years with fewer than 10 firms are dropped.

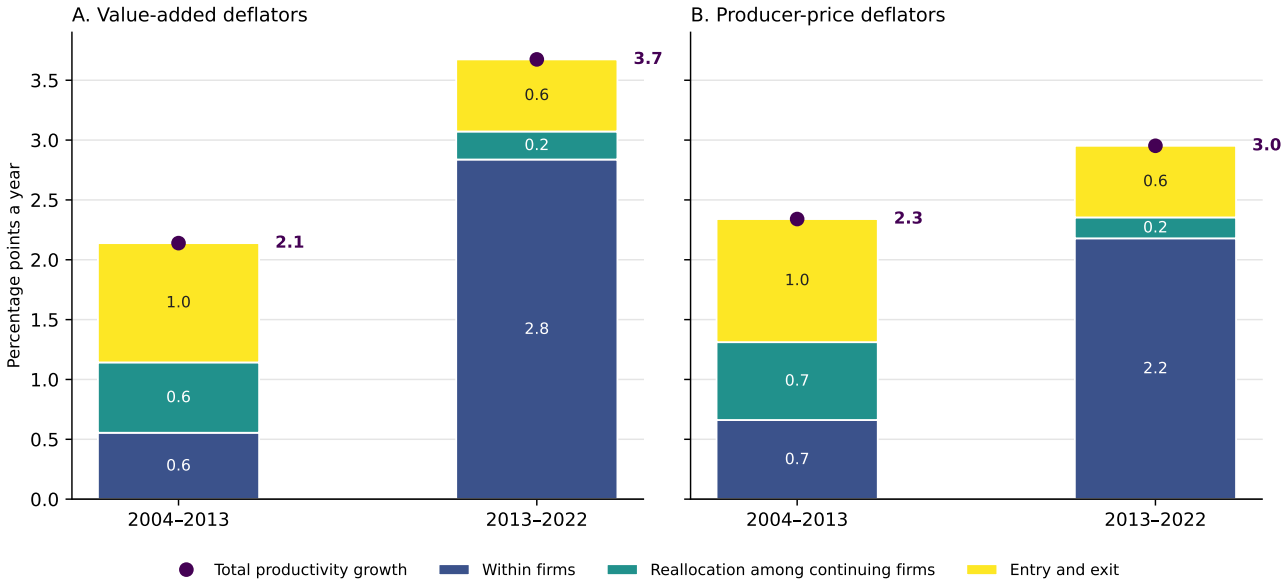


Figure 9: Exercise 2: all components of productivity growth before and after 2013, two price indices

Notes: Total productivity growth and its components (within firms, reallocation among continuing firms [between + cross], entry and exit), in percentage points a year, as in Table 1, 2004–2013 and 2013–2022. Panel A: industry value-added deflators from the national accounts (Eurostat, 2026). Panel B: industry producer-price deflators.

Levels. In the main sample, the efficiency gap averages 52% of value added in 2004–2010, 56% in 2010–2016, 58% in 2016–2019 and 62% in 2016–2023, with a peak of 68% in 2021. The periods share their boundary years (2010 and 2016). Among firms with 20 or more employees in CompNet industries, it rises from 42% to 46% between 2004–2010 and 2016–2023. In manufacturing, it stays at 30–32% in every period. Within sectors, the gap (in log points) rose between 2004–2010 and 2016–2023 in construction (from 38 to 67), utilities (40 to 57), information and communication (30 to 42), accommodation and food services (35 to 44), professional services (46 to 52), real estate (77 to 91) and trade (60 to 68; within trade, wholesale rose from 69 to 81 and motor trade from 46 to 54, while retail hardly changed, 41 to 42). It fell slightly in transport (60 to 58) and administrative services (39 to 37). Real estate has the largest gap but only 2% of value added, so it does not drive the aggregate.

Table 8: Efficiency gap, period averages

	2004–2010	2010–2016	2016–2019	2016–2023
Market economy, 5+ employees (main)	52.0	56.5	57.6	61.6
CompNet industries, 20+ employees	42.2	45.9	43.5	46.2
Manufacturing, 5+ employees	32.3	31.3	29.7	31.6
Market economy, $\sigma = 5$	74.9	85.0	87.8	97.3
Market economy, capital cost without imputed financing cost	43.0	45.3	46.7	50.0

Notes: Annual efficiency gaps, % of value added, averaged over the years shown; the periods share their boundary years, and 2016–2019 is part of 2016–2023. First three rows: main specification ($\sigma = 3$; capital cost including the imputed financing cost). The last two rows vary σ and the capital cost (without the imputed financing cost). 2023 is provisional.

From the efficiency gap to the reserve. The benchmark used in the main text is $(1 + g_{\text{later}})/(1 + g_{2004–10}) - 1$: the proportional increase in the later period's output if the model's efficiency ratio returned to its 2004–2010 average, applying the 2004–2010 efficiency ratio to the later period's efficient output. In the main specification, it is 6.3% of value added with 2016–2023 as the later period (6.2% when computed from period averages of the log gain). This is a difference in the level of output, not an annual growth rate, and it neither restores the historical allocation firm by firm nor models the transition.

Expressing it relative to GDP requires a value-added base. The trimming removes about 3% of firms but 31–44% of value added, depending on the year, because the top tail of the A_i deviation contains many of the largest firms. In

2023, the calculation keeps 60,935 of 62,857 firms and HUF 20,113 billion of HUF 29,979 billion of value added. Applied to the firms kept, 6.3% is HUF 1,264 billion, or 1.7% of 2023 GDP. Applied to all firms with 5 or more employees, which assumes that the excluded firms would gain in the same proportion, it amounts to HUF 1,884 billion, or 2.5% of GDP. Most of the trimmed value added is in the top tail of physical productivity, among the firms the model says should grow most; the effect of including them on the gap cannot be signed in general (Table 9 shows the result without trimming).

Sensitivity. Table 9 reports the benchmark and its GDP share across variants. With 2016–2023 as the later period, it ranges from 0.8% to 5.1% of GDP, and exceeds 1% of GDP in every variant except the 2% trim applied to the firms kept. With only the pre-pandemic years 2016–2019 as the later period, it is smaller and less robust, at 0.2–2.0% of GDP, because the efficiency gap rose most in 2020–2023. Without trimming, a few extreme firms push the level of the efficiency gap to about 200%. The benchmark for 2016–2023 is then similar (6.7%), but for 2016–2019 it falls to 0.6% of value added. With $\sigma = 5$, the gap rises from 75% to 97% between 2004–2010 and 2016–2023, a rise about twice as large as in the main specification, and the benchmark is 12.8% of value added; $\sigma = 3$ is therefore the conservative choice. Without the imputed financing cost, the gap is lower (43% to 50%) and rises by seven points, against ten in the main specification. For firms with 20 or more employees in CompNet industries, the benchmark is 2.8% of their value added; in manufacturing alone, it is about zero (−0.5%).

Table 9: The return-to-2004–2010 benchmark across specifications, as a share of value added and of GDP

	2016–2023 vs 2004–2010			2016–2019 vs 2004–2010		
	Benchmark	% of GDP		Benchmark	% of GDP	
Variant	% of VA	kept firms	all 5+ firms	% of VA	kept firms	all 5+ firms
Trim 1% (main)	6.3	1.7	2.5	3.7	1.0	1.5
No trimming	6.7	2.6	2.7	0.6	0.2	0.3
Trim 0.5%	8.7	2.6	3.5	5.1	1.5	2.0
Trim 2%	3.6	0.8	1.4	1.5	0.4	0.6
$\sigma = 5$	12.8	–	5.1	–	–	–
Capital cost without financing cost	4.9	–	1.9	–	–	–

Notes: Benchmark: $(1 + g_{\text{later}})/(1 + g_{2004–10}) - 1$, from period averages of the annual gain, main sample unless stated. Share of 2023 GDP (HUF 75,218 billion), applied to the 2023 value added of the firms kept under each rule or of all market-economy firms with 5+ employees (HUF 29,979 billion). Trim: percentage removed at each tail of both deviations. The $\sigma = 5$ and capital-cost variants use the 1% trim, and only the all-firm base is shown for them. Without trimming, firms are still dropped by the positivity filters and the 10-firm minimum, so the two bases differ slightly.

Limitations. The level of the efficiency gap depends on the model (CES demand with a common σ, Cobb–Douglas technology with industry factor shares), on two-digit industries and on trimming. Hungarian factor shares may themselves reflect distortions, which is why Hsieh and Klenow use US shares. Capital is measured at book value. The benchmark does not identify which policies created the wedges, or what a particular reform would deliver. The GDP share relates a change in the value added of market-economy firms with 5 or more employees to total GDP, and so assumes no gain in the rest of the economy. For these reasons, the main text reports an order of magnitude rather than a point estimate: a one-off rise in the level of output of about 1–2% of GDP (0.2–0.4 percentage points of extra growth a year if reached over five years), within a range of 0.8–5.1% of GDP across the variants in Table 9.

Sector decomposition and dispersion. Since $\ln(Y^{\text{eff}}/Y) = \sum_s \theta_s \ln(TFP_s^{\text{eff}}/TFP_s)$ is additive across industries, it can be grouped into eleven sectors. Between 2004–2010 and 2016–2023, the aggregate rises from 41.9 to 47.9 log points: +6.4 from larger gaps within sectors at initial shares, and −0.4 from changes in sector shares (with symmetric weights at the two-digit level, +6.8 and −0.8). The standard deviation of logs within industry-years, averaged over 2004–2010, 2010–2016 and 2016–2023, is 0.51, 0.52 and 0.52 for $TFPR$; 1.45, 1.58 and 1.57 for the return on capital; and 0.55, 0.53 and 0.53 for the return on labour. The dispersion of returns on capital rose early and then stayed high, while the efficiency gap kept rising. The later rise in the gap therefore reflects how wedges line up with firm productivity, not how widely they are spread.

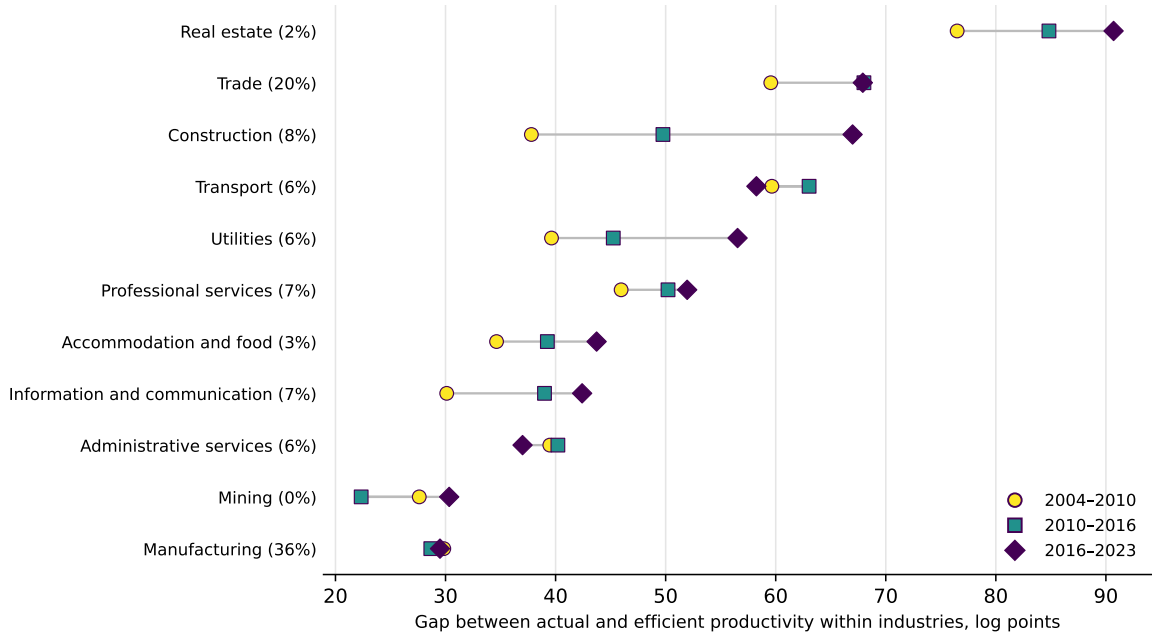


Figure 10: Exercise 3: efficiency gap by sector

Notes: Value-added-weighted average of $100 \ln(TFP^{\text{eff}}/TFP)$ across the two-digit industries in each sector, in log points; main sample, period averages. Sector shares in value added in 2016–2023 are in brackets.

Poland. Gradzewicz (2026) applies the same method to Polish firms with 10 or more employees and finds that the efficiency gap rose from close to 40% in the 1990s to 70–75% in the early 2020s. The samples and assumptions differ from ours, so the levels are not comparable.

H Replication and checks

The results come from three automated pipelines (built with make, using Stata 18 and Python 3), in the folders `tfp/misallocation` (Exercise 1), `tfp/reallocation` (Exercise 2) and `tfp/hsieh-klenow` (Exercise 3). Each has a README, a decision log and its aggregated outputs. All tables are generated from the output files by a single script; numbers quoted in the text were checked against those files but entered by hand (work in progress, item (vi)). The dispersion series (Exercise 1) and the Hsieh–Klenow efficiency gaps (Exercise 3) were independently re-implemented in Python, matching to within 0.0003 and to machine precision, respectively. The decomposition of Exercise 2 was not re-implemented; its terms add up exactly, which the code verifies (Appendix E). Two independent reviews of earlier versions, and our responses to them, are filed with the code. The firm-level data are licensed and remain in the research environment; only aggregates are published.

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